Sthyra CRM — System Architecture

Complete Service Inventory, AWS Mapping, and Flow Diagrams

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Executive Summary

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Executive Summary

This document is the **system architecture** for Sthyra CRM — the AWS deployment target that complements the Phase 0 report. While the Phase 0 report documents *what* was built (the foundation), this document specifies *how* the full 13-product platform runs in production on AWS.

The architecture has three planes — **edge**, **control**, **data** — and three layers of trust — **public internet**, **authenticated API surface**, **internal service mesh**. Every service is deployed as a container on EKS, fronted by an Envoy-based API gateway, with RDS Postgres for the system of record, S3 for media, ElastiCache Redis for sessions and cache, and a dedicated GPU node group for the ML/3D pipeline.

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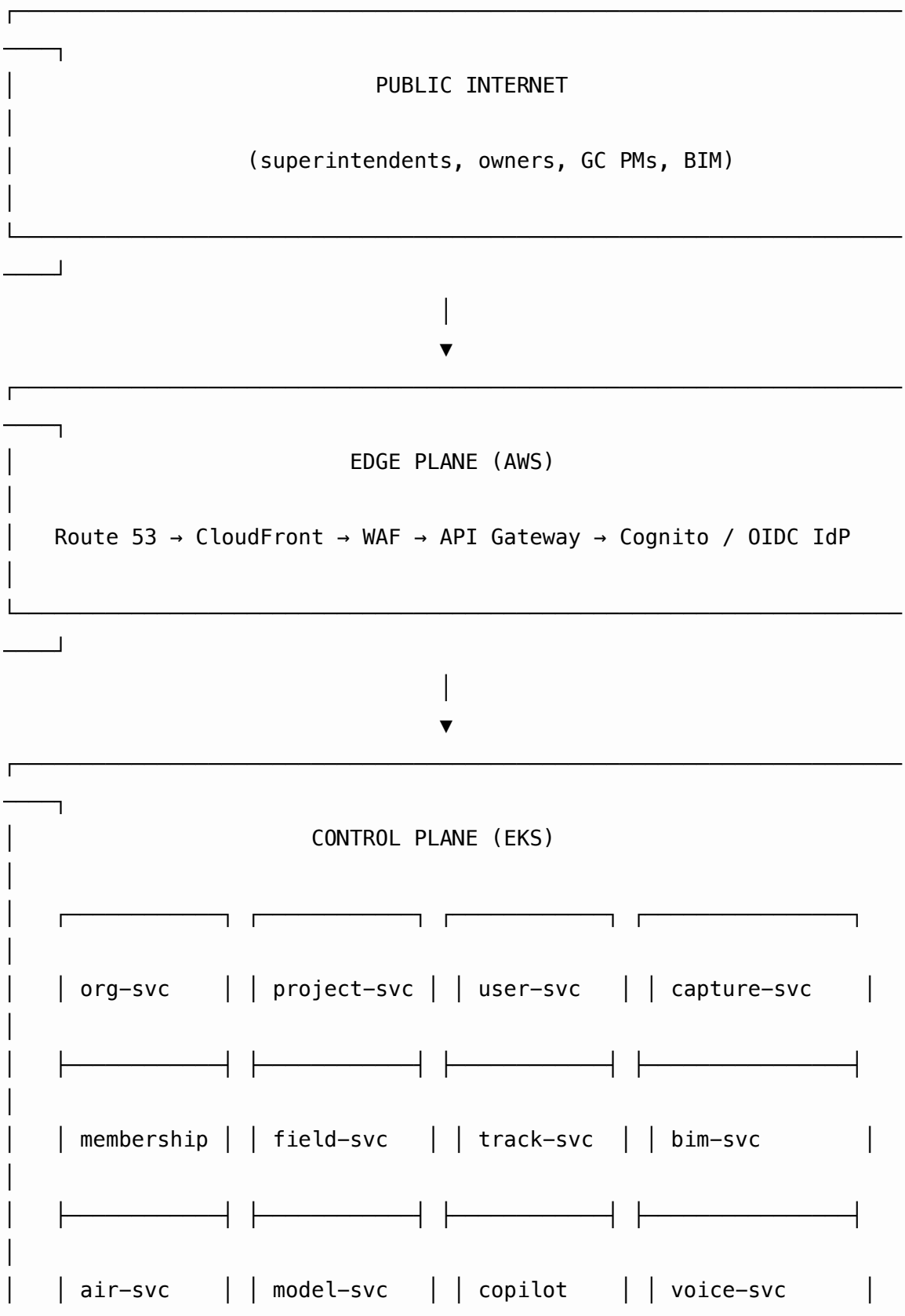
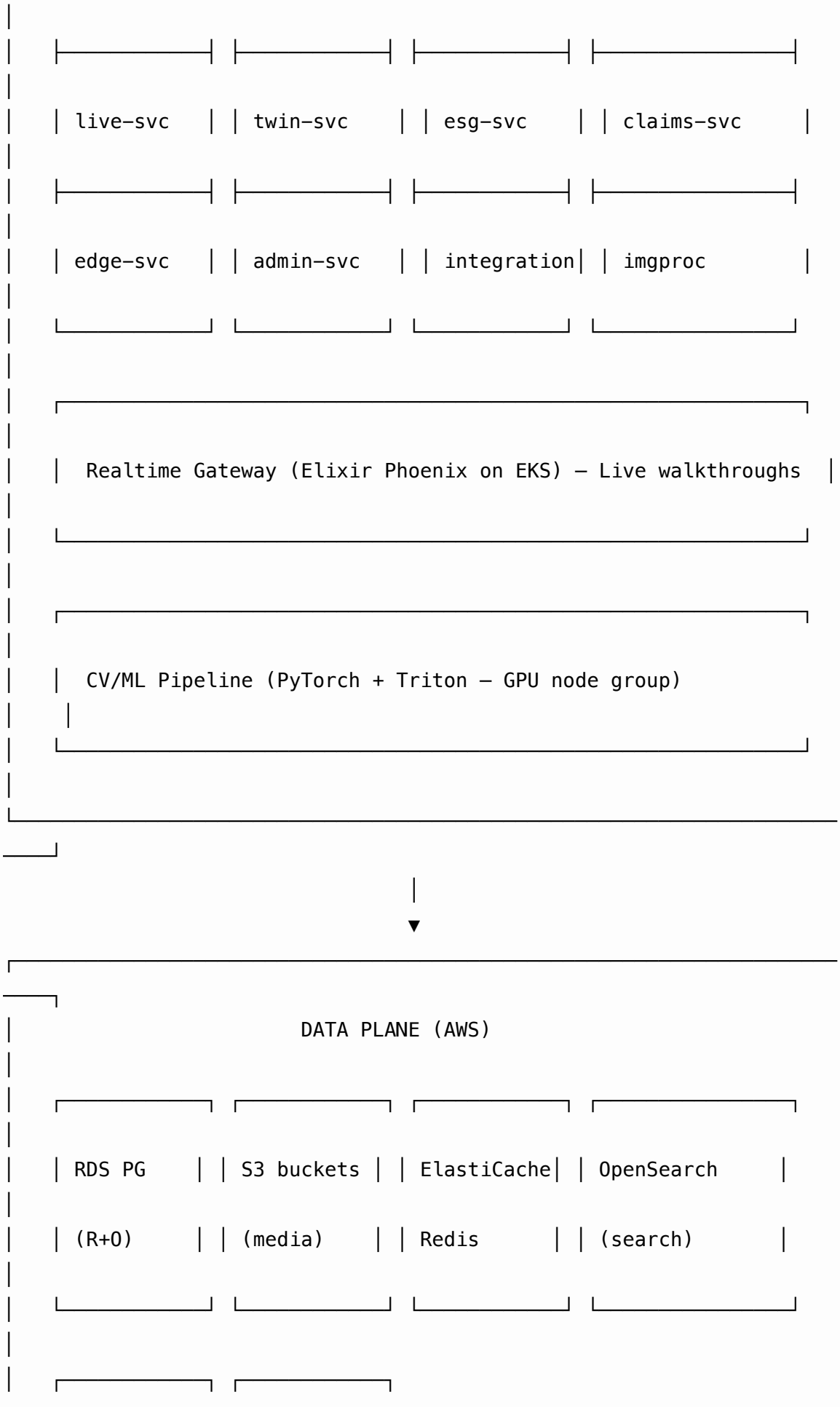
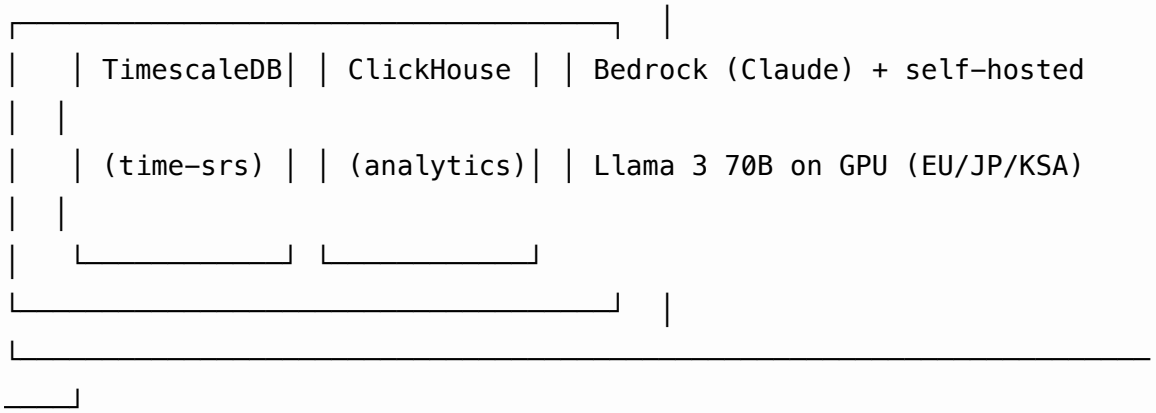
1. System Overview

Sthyra CRM is a multi-tenant SaaS platform for construction visual intelligence. Three planes:




live-svc
twin-svc
esg-svc
claims-svc
edge-svc
admin-svc
integration
imgproc
Realtime Gateway (Elixir Phoenix on EKS) — Live walkthroughs
CV/ML Pipeline (PyTorch + Triton — GPU node group)
DATA PLANE (AWS)
RDS PG (R+O)
S3 buckets (media)
ElastiCache Redis
OpenSearch (search)



2. AWS Service Inventory

Every AWS service used by Sthyra CRM, with the Sthyra CRM component it serves and the closest non-AWS alternative annotated for clarity.

Layer	AWS Service	Sthyra CRM Component	Notes
DNS	Route 53	Apex + per-region failover	Health-checked alias records to CloudFront
CDN	CloudFront	Static assets, marketing site, 360° tile streaming	OAI to S3; Lambda@Edge for auth checks
WAF	AWS WAF	Edge protection	OWASP Top 10 managed rules + custom rate-limit rules
DDoS	AWS Shield Standard + Advanced	Edge protection	Automatic
API Gateway	AWS API Gateway (REST)	Public REST surface	VPC integration; integrates with WAF
Load Balancer	Network Load Balancer (NLB)	Internal service ingress	TLS termination at Ingress-NGINX

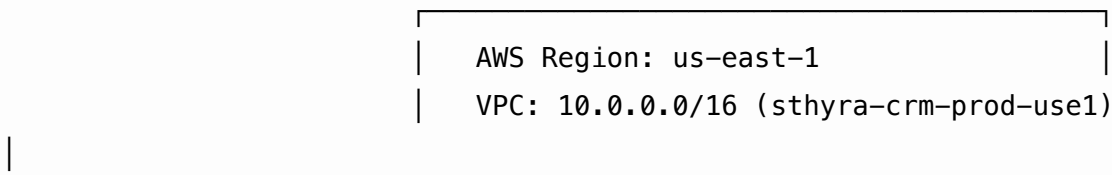
Layer	AWS Service	Sthyra CRM Component	Notes
Service Mesh	App Mesh (Envoy)	East-west traffic	mTLS via SPIFFE; 2.5% sampling to X-Ray
Compute	EKS (Kubernetes)	All services	Karpenter for node autoscaling
GPU	EC2 P5 / G5 instances	CV/ML pipeline, 3DGS, ME	Karpenter MIG profiles for fine-grained scheduling
Serverless	Lambda	OIDC callbacks, S3 event triggers, ConMon	1-2 GB / 30s timeout
Database	RDS for PostgreSQL (Aurora)	System of record	Multi-AZ, encryption at rest, KMS-managed
Time-series	Timescale Cloud (managed) on RDS	Capture telemetry, sync events	Hypertable partitioning
OLAP	ClickHouse Cloud (or Redshift Serverless)	Analytics dashboards	100× cheaper than Postgres for scans
Search	OpenSearch Service	Federated search	Per-tenant index aliases
Vector	Aurora pgvector (Phase 1) → OpenSearch k-NN (Phase 2)	Copilot RAG	Hybrid BM25 + dense
Cache	ElastiCache for Redis	Sessions, rate-limit, idempotency, presence	Cluster mode, AOF persistence
Queue	SQS + Kinesis Data Streams	Ingestion, telemetry, webhooks	Standard + FIFO; Kinesis for backpressure

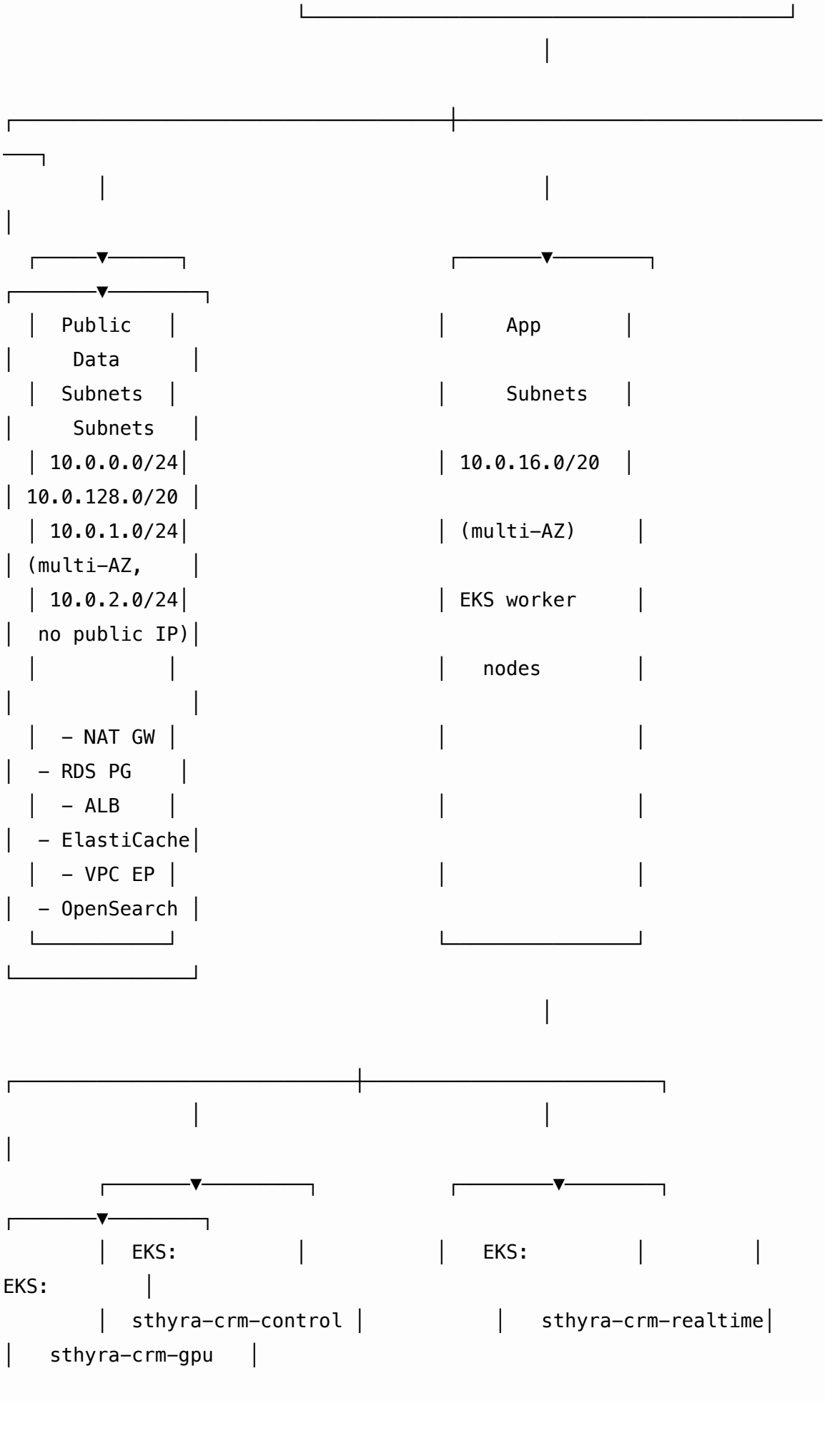
Layer	AWS Service	Sthyra CRM Component	Notes
Object storage	S3	Media (raw 360, frames, tiles, BIM, drone)	Lifecycle: hot → IA → Glacier
Block storage	EBS gp3	Postgres data, Redis	Encrypted, gp3 for cost
CDN origin	S3 + CloudFront Origin Shield	Tile streaming	Per-tenant signed URLs
Secrets	AWS Secrets Manager	DB creds, OIDC client secrets	Auto-rotation 90d
KMS	AWS KMS + CloudHSM	All encryption-at-rest	HSM-backed root for FedRAMP-High
IAM	AWS IAM + IRSA	Service-to-AWS auth	SPIFFE/SVID → IAM roles via OIDC
Identity	Amazon Cognito (user pools) + Auth0 / Okta	Customer IDP	Federated SSO per tenant
Logging	CloudWatch Logs + S3	Structured logs	OTel → Firehose → S3 + OpenSearch
Metrics	CloudWatch Metrics + Prometheus	Service metrics	1s resolution for control-plane alerts
Tracing	AWS X-Ray + OpenTelemetry	Distributed tracing	2.5% sampling default
Dashboards	CloudWatch Dashboards + Grafana	Ops + product dashboards	Grafana for SLOs
Alerts	CloudWatch Alarms + PagerDuty	On-call paging	SEV1-4 severity model
Secrets	AWS Secrets Manager	Cross-service secrets	Per-secret IAM policy

Layer	AWS Service	Sthyra CRM Component	Notes
Containers	ECR	Service images	Lifecycle policies: keep last 30
CI/CD	GitHub Actions + ArgoCD	Build + deploy	GitOps on every commit
IaC	Terraform + AWS CDK	Infrastructure	Per-region state files in S3
Key mgmt	AWS KMS + CloudHSM	CMK per tenant, per-region	HSM-backed root for FedRAMP-High
WAF	AWS WAF	Layer 7 protection	Managed rule sets + custom
DNS	Route 53	Geo-routing for FedRAMP-boundary	Latency-based records
Cost	AWS Cost Explorer + CUR	Per-tenant cost attribution	Tag-based allocation
DR	AWS Backup + S3 Cross-Region Replication	RPO ≤ 15 min / RTO ≤ 4 h	Daily drills
Audit	AWS CloudTrail + S3 Object Lock	Compliance evidence	7-year retention
Secrets	AWS Secrets Manager	Per-tenant credentials	Cross-region replication

3. Network Topology

Each AWS region deploys the same VPC layout. The diagram below is for `us-east-1`; `eu-west-1`, `ap-southeast-2`, `ap-northeast-1`, and `ksa-central-1` mirror it with regional IP ranges.




Public Subnets
10.0.0.0/24
10.0.1.0/24
10.0.2.0/24
(multi-AZ, no public IP)
- NAT GW
- RDS PG
- ALB
- ElastiCache
- VPC EP
- OpenSearch
Data Subnets
10.0.128.0/20
(multi-AZ)
App Subnets
10.0.16.0/20
(multi-AZ)
EKS worker nodes
EKS: sthyra-crm-control
EKS: sthyra-crm-realtime
EKS: sthyra-crm-gpu

```
        │  (stateless    │           │   (Phoenix       │        │
(P5/G5 nodes)│
        │   services)    │           │    Channels)     │        │
│
        └────────────────┘           └──────────────────┘
└────────────────┘

INTER-REGION:
- Transit Gateway peering: us-east-1 ↔ us-west-2 ↔ eu-west-1
- GovCloud (us-gov-east-1) is ISOLATED — no transit peering to
commercial
- S3 Cross-Region Replication: raw-360 → eu-west-1 (active-active)
- RDS Cross-Region Read Replica: us-east-1 → us-west-2 (warm standby)
- Route 53 latency-based routing across regions
```

3.1 Subnet design rationale

Subnet tier	CIDR	Purpose	Ingress
Public	/24 × 3 AZs	NAT gateway, ALB, VPC endpoints	Internet Gateway
App	/20 × 3 AZs	EKS worker nodes (stateless services)	NAT → Internet for outbound
Data	/20 × 3 AZs	RDS Aurora, ElastiCache, OpenSearch	No internet; VPC endpoints only
GPU	/20 × 3 AZs	EKS GPU node group (P5/G5)	App subnets + EFA
Spare	—	Reserved for VPC expansion	Future

3.2 Security groups (per service)

```
┌──────────────────────────────────────────────────────────────────────
────┐
│  sthyra-crm-control-sg
│
│  Ingress: 443/ALB-sg, 10250/65000-65535/control-plane-sg
```

```
│
│  Egress:  5432/data-sg, 6379/redis-sg, 443/0.0.0.0/0
│
└──────────────────────────────────────────────────────────────────────────────
────┘

┌──────────────────────────────────────────────────────────────────────────────
────┐
│  sthyra-crm-realtime-sg (Phoenix Channels)
│
│  Ingress: 443/ALB-sg, 4000-4010 (WebSocket)
│
│  Egress:  6379/redis-sg
│
└──────────────────────────────────────────────────────────────────────────────
────┘

┌──────────────────────────────────────────────────────────────────────────────
────┐
│  sthyra-crm-gpu-sg (CV/ML pipeline)
│
│  Ingress: 10250/control-plane-sg
│
│  Egress: 5432/data-sg, 443/S3-vpce, 443/Bedrock-vpce
│
└──────────────────────────────────────────────────────────────────────────────
────┘

┌──────────────────────────────────────────────────────────────────────────────
────┐
│  sthyra-crm-data-sg (RDS, ElastiCache, OpenSearch) — NO INGRESS
│
│  Ingress: 5432/control-sg, 5432/gpu-sg, 6379/control-sg,
9200/control │
└──────────────────────────────────────────────────────────────────────────────
────┘
```

4. Service Inventory

Every Sthyra CRM service, its AWS deployment shape, and the AWS services it depends on. **Phase 0 services** are built; **Phase 1+ services** are specified in the Phase 0 report.

4.1 Phase 0 services (built; running on EKS)

Service	Image	EKS Cluster	Replicas (min/max)	Dependencies
`org-service`	`org-service:v0.1.0`	sthyra-crm-control	2/6	RDS Postgres, Secrets Manager
`project-service`	`project-service:v0.1.0`	sthyra-crm-control	2/6	RDS Postgres
`user-service`	`user-service:v0.1.0`	sthyra-crm-control	2/6	RDS Postgres, Amazon Cognito
`membership-service`	`membership-service:v0.1.0`	sthyra-crm-control	2/6	RDS Postgres, user-service
`dashboard` (Next.js)	`dashboard:v0.1.0`	sthyra-crm-control	2/10	EFS for `.next/cache`, all services

4.2 Phase 1 services (specification)

Service	Purpose	AWS Resources	GPU?
`copilot-service`	LLM gateway, function-calling, RAG	EKS control plane + Bedrock (Claude) + Aurora pgvector	No
`capture-service`	360° video ingestion, upload-session orchestration	EKS control plane + S3 + SQS + Step Functions	No

Service	Purpose	AWS Resources	GPU?
`imgproc-service`	Spatial AI: COLMAP/GLOMAP, OpenMVS, 3D Gaussian Splatting, SAM-2, DINOv2	EKS GPU node group (P5.48xlarge) + FSx for Lustre	Yes — P5/H100
`bim-service`	BIM ingest (IFC, Revit, NWD), tessellation, clash detection	EKS control plane + S3 + 3D Tiles hosting	No
`field-service`	Field notes, issues, dictation, sketches	EKS control plane + RDS Postgres + S3 (attach)	No
`track-service`	Progress tracking: BIM-vs-reality alignment, percent-complete	EKS control plane + TimescaleDB + ClickHouse	No
`model-service`	3D model serving, MeshLab glTF	EKS control plane + S3 + CloudFront	No
`air-service`	Drone coordination, LAANC, BVLOS-aware flight logs	EKS control plane + S3 + Kinesis (telemetry)	No
`admin-service`	Org/project management, billing, audit log	EKS control plane + RDS Postgres + Athena	No
`integration-service` (×14)	Connectors: Procore, ACC, BIM 360, P6, Slack, Teams, etc.	EKS control plane + SQS + Secrets Manager	No

4.3 Phase 2 services

Service	AWS Resources
`voice-service`	Lambda + Transcribe + Polly (realtime)
`live-service`	EKS Phoenix + IVS (real-time streaming) + Kinesis Video
`twin-svc`	EKS + IoT TwinMaker + S3
`esg-svc`	EKS + Bedrock (Anthropic) for report generation
`claims-svc`	EKS + QLDB (immutable ledger) + S3

4.4 Shared infrastructure services

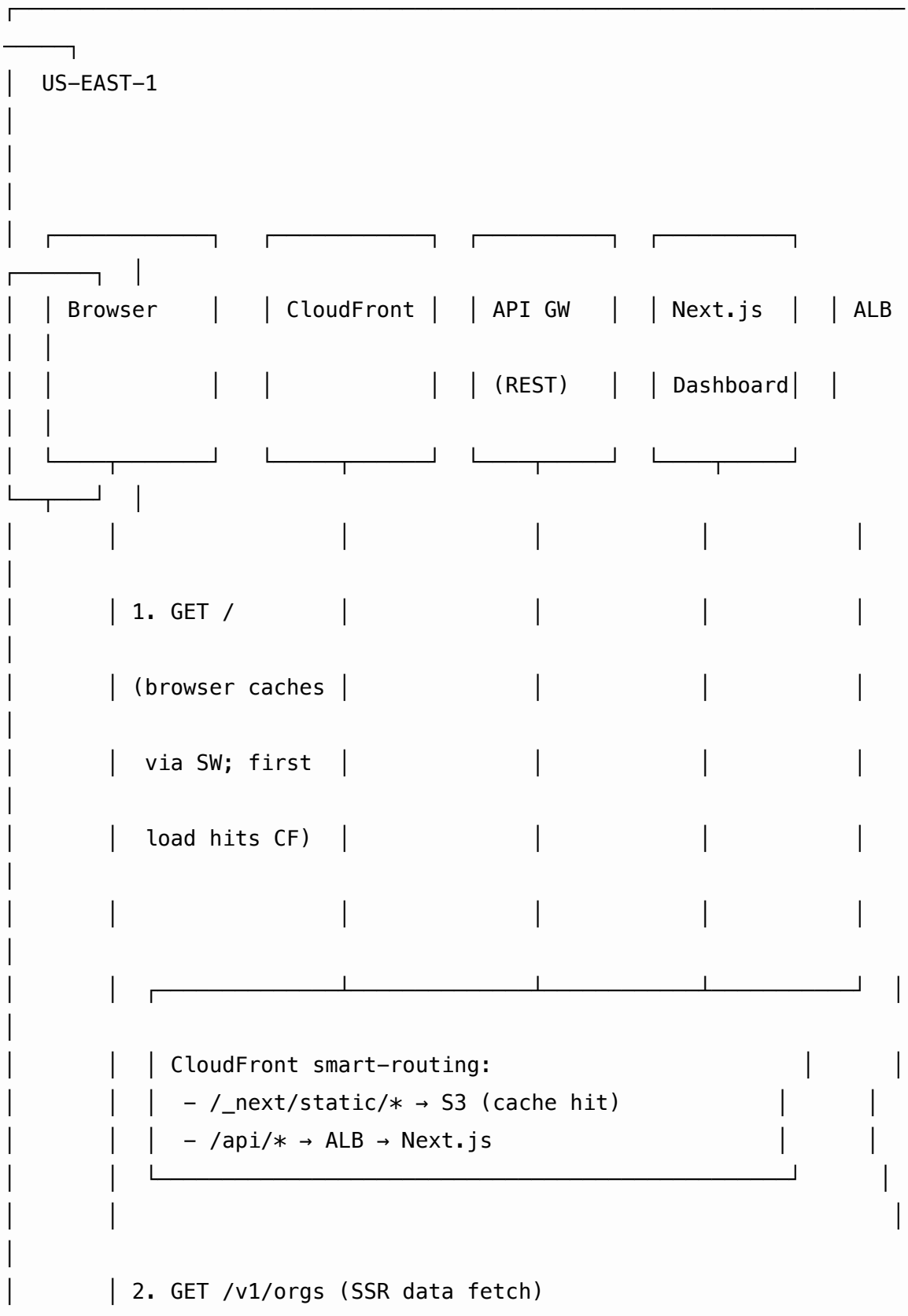
Service	AWS Resource	Purpose
`realtime-gateway`	EKS Phoenix Channels on `sthyra-crm-realtime` cluster	WebSocket presence + live walkthroughs
`pipeline-orchestrator`	Step Functions + Lambda	Coordinates Spatial AI pipeline
`tile-server`	CloudFront + Lambda@Edge	Streams 3D tiles to browsers
`auth-broker`	EKS + Amazon Cognito + Lambda	OAuth2/OIDC + SAML
`audit-collector`	Kinesis Firehose → S3 + Athena	Compliance evidence
`cost-collector`	AWS Cost and Usage Reports → S3 → Athena	Per-tenant cost allocation
`event-bus`	Amazon EventBridge	Cross-service domain events

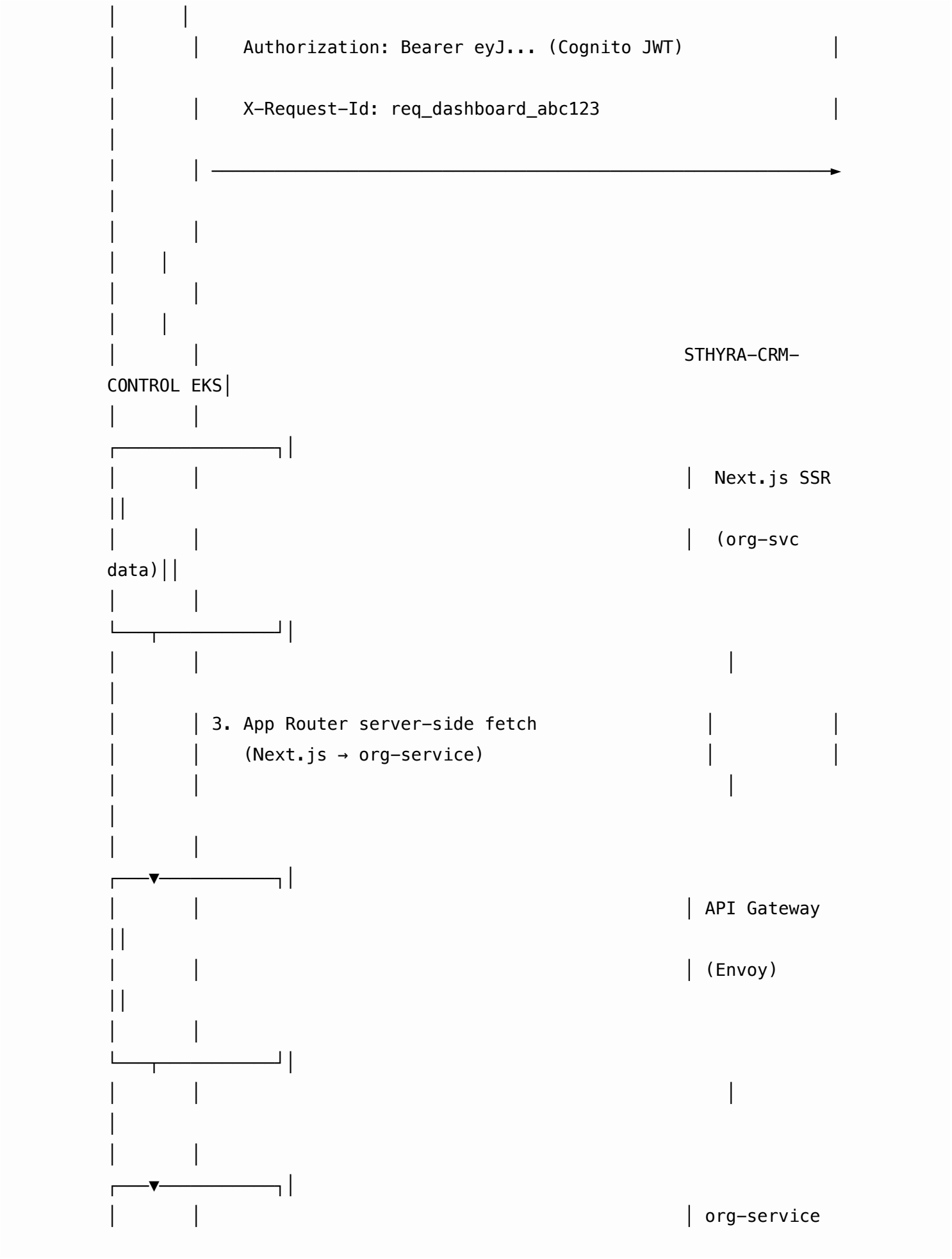
5. Flow Diagrams

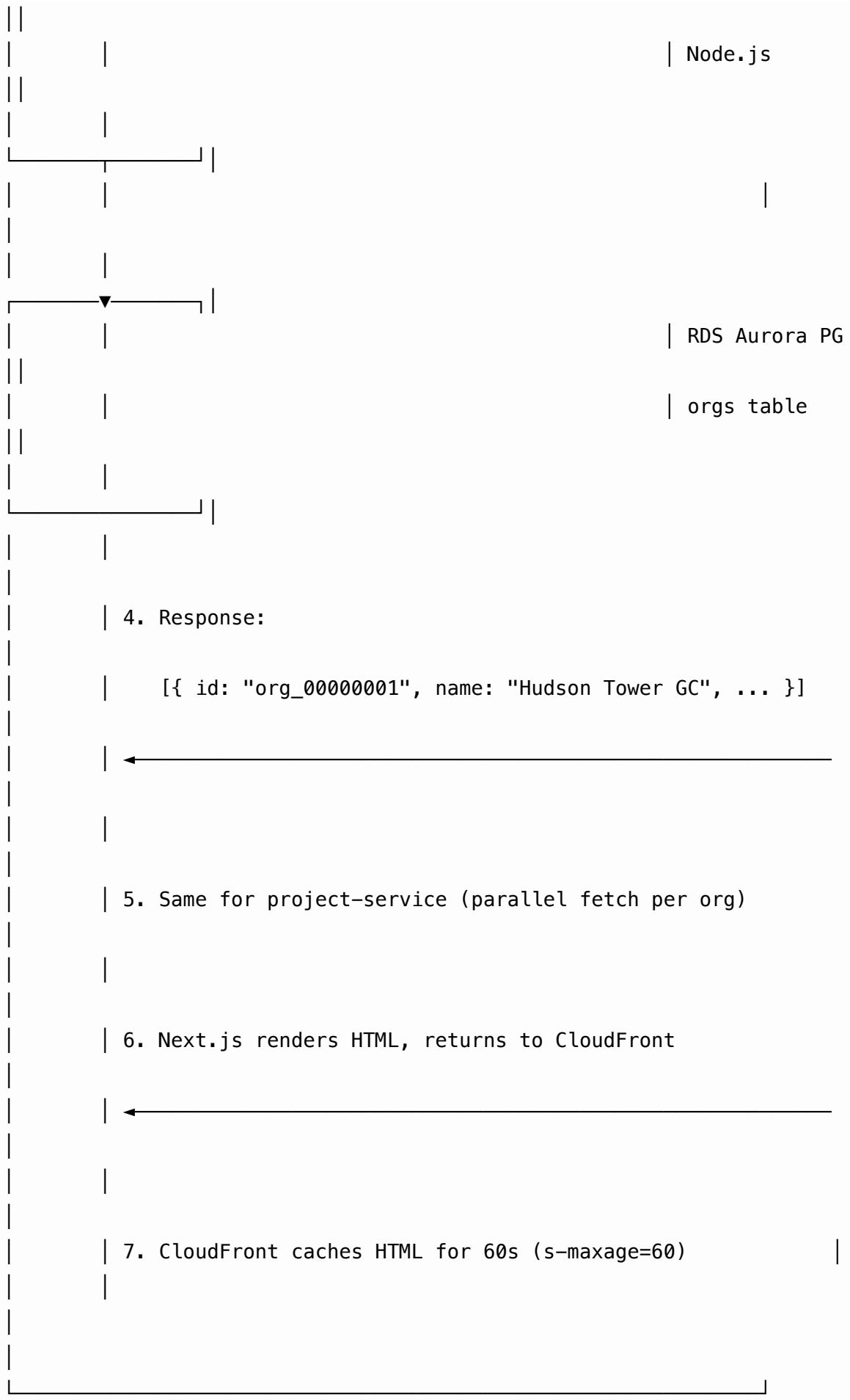
The flow diagrams below show the **major request paths** through the system. Each diagram names the AWS service, the Sthyra CRM service, and the data exchanged at each step.

5.1 User opens the dashboard

A VDC PM opens `https://app.sthyra-crm.dev` and lands on the homepage showing org/project rollups.




Authorization: Bearer eyJ... (Cognito JWT)
X-Request-Id: req_dashboard_abc123
STHYRA-CRM-CONTROL EKS
Next.js SSR (org-svc data)
3. App Router server-side fetch
(Next.js → org-service)
API Gateway (Envoy)
org-service

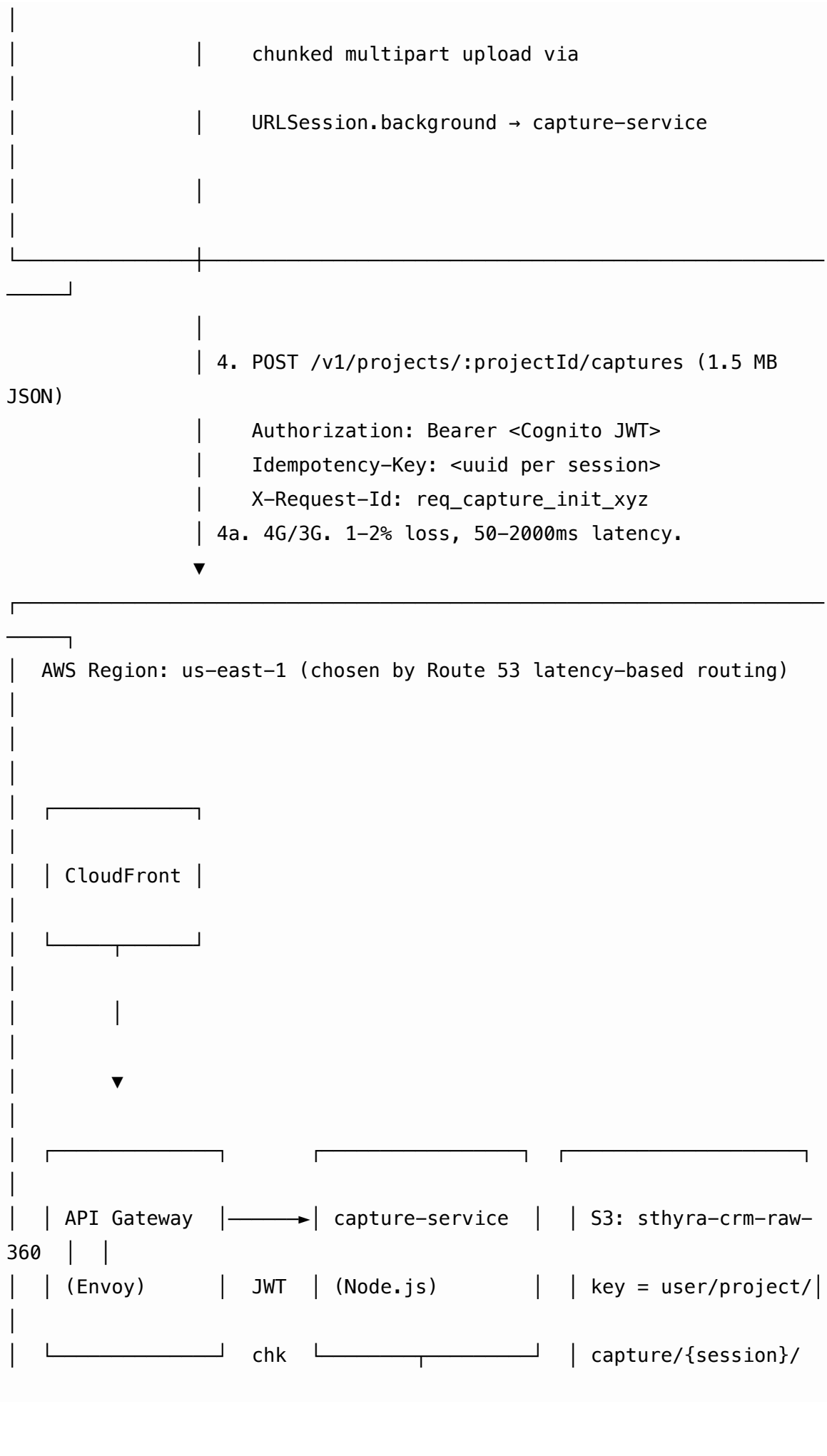

Node.js
RDS Aurora PG
orgs table
4. Response:
[{ id: "org_00000001", name: "Hudson Tower GC", ... }]
5. Same for project-service (parallel fetch per org)
6. Next.js renders HTML, returns to CloudFront
7. CloudFront caches HTML for 60s (s-maxage=60)

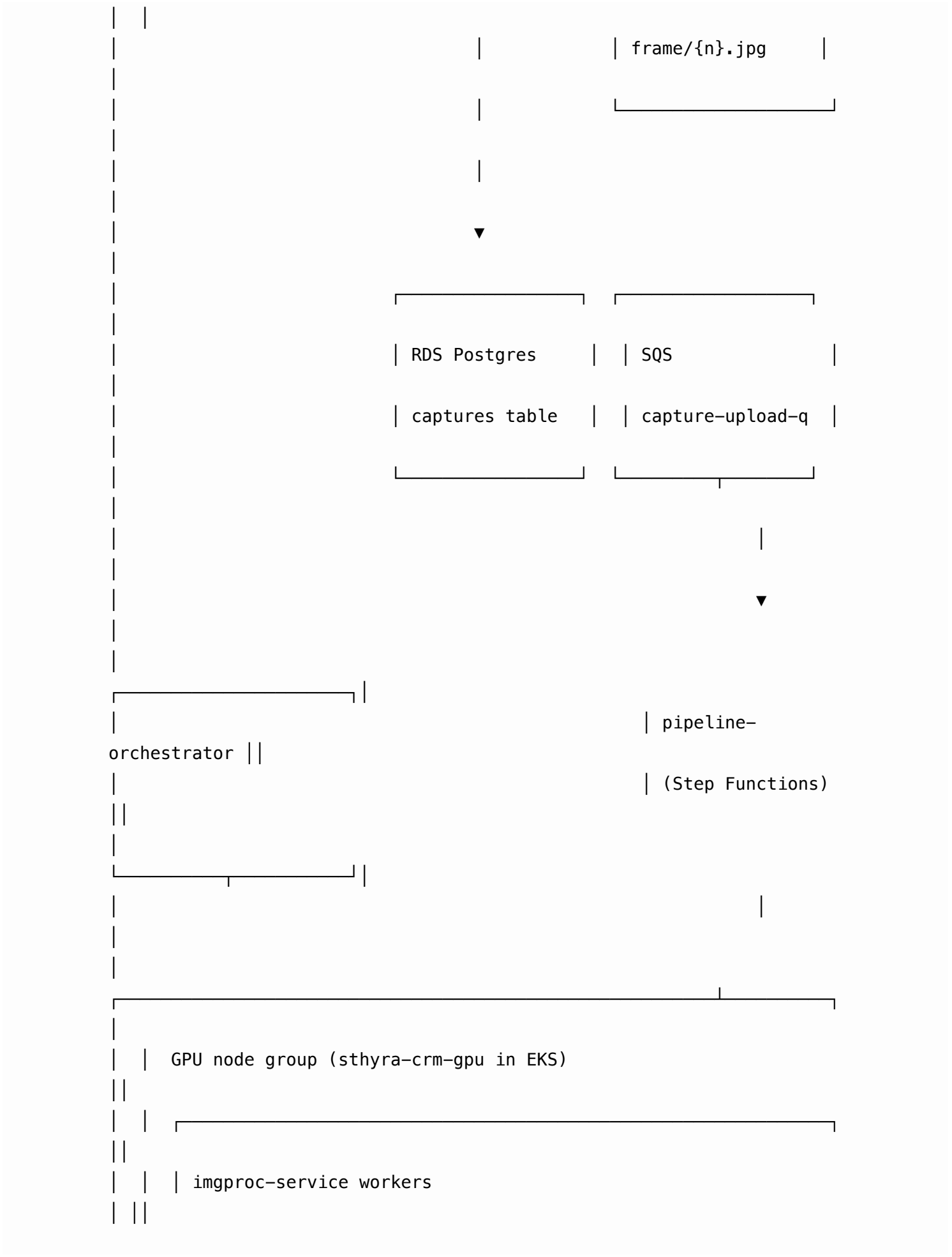
Key properties: - All CloudFront → ALB traffic is HTTPS (TLS 1.3) - API Gateway validates the Cognito JWT on every request - `x-request-id` is propagated through every hop (per `@sthyra-crm/observability`) - Next.js SSR fetch uses `force-dynamic` so the home page is always fresh - HTML is CDN-cached for 60s; API responses are not cached

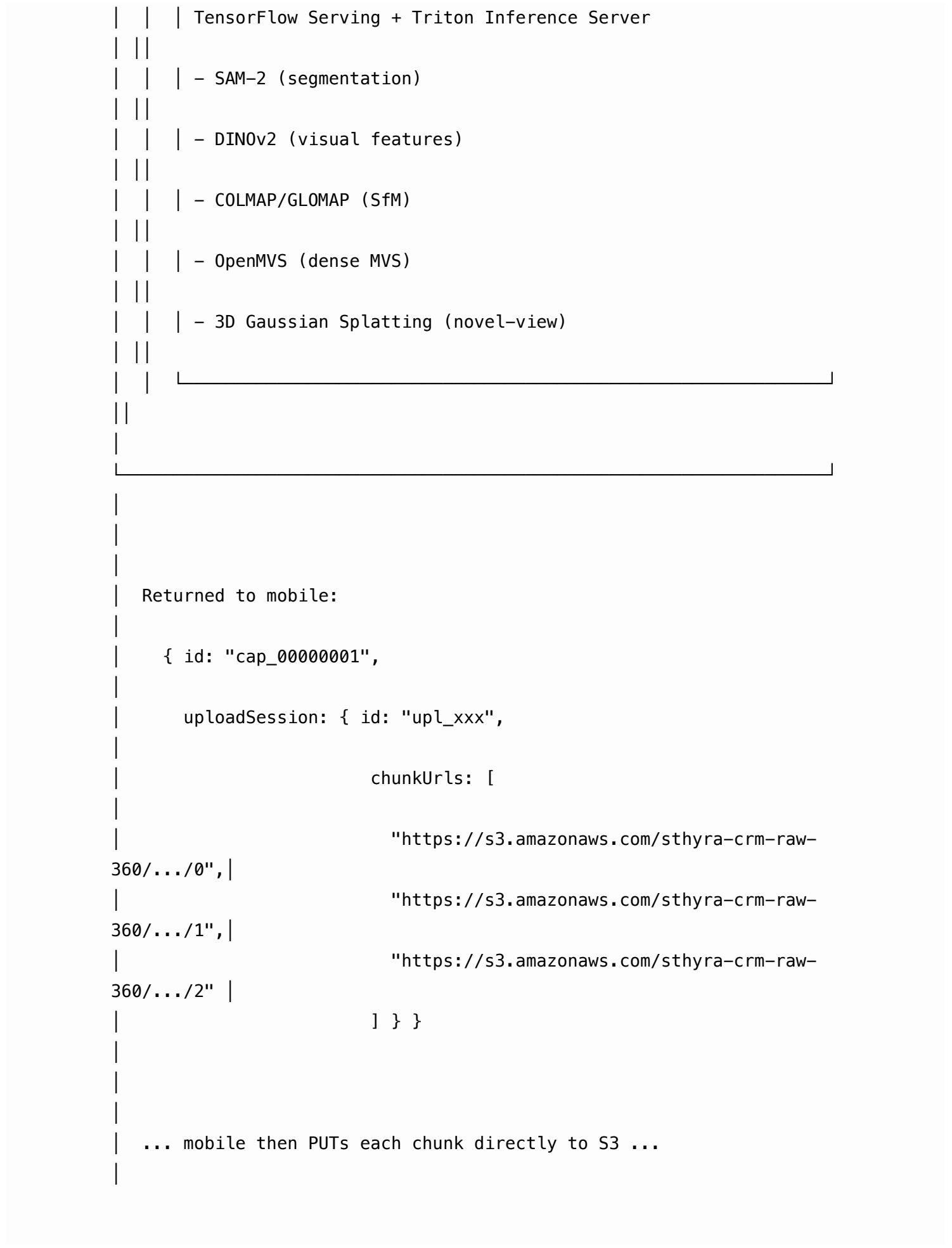
5.2 Field user starts a 360° capture

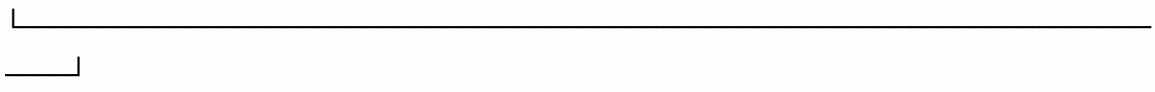
A superintendent in the field, on iPhone with Bluetooth-paired Insta360 X4, taps “Start walk” in the mobile app.

```
┌──────────────────────────────────────────────────────────────────────────
─────┐
│  CONSTRUCTION SITE (3G/4G/wifi)
│
│
│
│  ┌──────────────────────────┐
┌──────────────────────────┐ │
│  │ iPhone + Insta360 X4    │               │ Base station / trailer
│ │
│  │ mobile-kmm (KMP)        │  Bluetooth     │ (rare; usually no)
│ │
│  └──────────────┬──────────┘
└──────────────────────────┘ │
│               │
│
│               │ 1. CaptureSession row inserted in
│
│               │    on-device SQLite (offline-first)
│
│               │
│
│               │ 2. AVCaptureSession records 8K equirectangular
frames  │
│               │    at 30fps; CoreMotion fuses IMU + GPS
│
│               │
│
│               │ 3. As frames are encoded, KMM sync engine begins
```


chunked multipart upload via
URLSession.background → capture-service
4. POST /v1/projects/:projectId/captures (1.5 MB JSON)
Authorization: Bearer <Cognito JWT>
Idempotency-Key: <uuid per session>
X-Request-Id: req_capture_init_xyz
4a. 4G/3G. 1–2% loss, 50–2000ms latency.
AWS Region: us-east-1 (chosen by Route 53 latency-based routing)
CloudFront
API Gateway
(Envoy)
JWT
chk
capture-service
(Node.js)
S3: sthyra-crm-raw-360
key = user/project/
capture/{session}/


frame/{n}.jpg
RDS Postgres
captures table
SQS
capture-upload-q
pipeline-
orchestrator
(Step Functions)
GPU node group (sthyra-crm-gpu in EKS)
imgproc-service workers

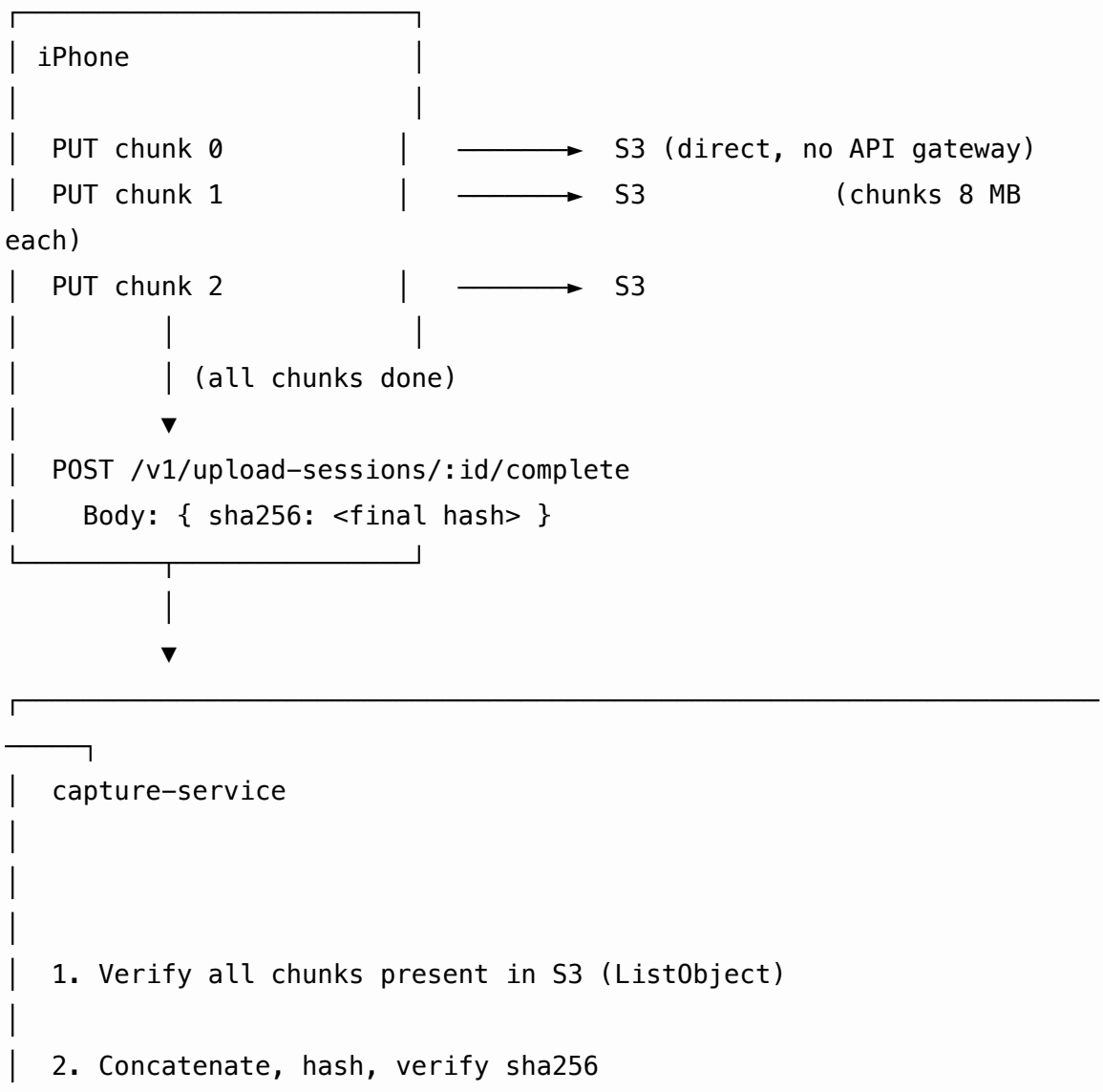

TensorFlow Serving + Triton Inference Server
- SAM-2 (segmentation)
- DINOv2 (visual features)
- COLMAP/GLOMAP (SfM)
- OpenMVS (dense MVS)
- 3D Gaussian Splatting (novel-view)
Returned to mobile:
{ id: "cap_00000001",
uploadSession: { id: "upl_xxx",
chunkUrls: [
"https://s3.amazonaws.com/sthyra-crm-raw-360/.../0",
"https://s3.amazonaws.com/sthyra-crm-raw-360/.../1",
"https://s3.amazonaws.com/sthyra-crm-raw-360/.../2"
] } }
... mobile then PUTs each chunk directly to S3 ...

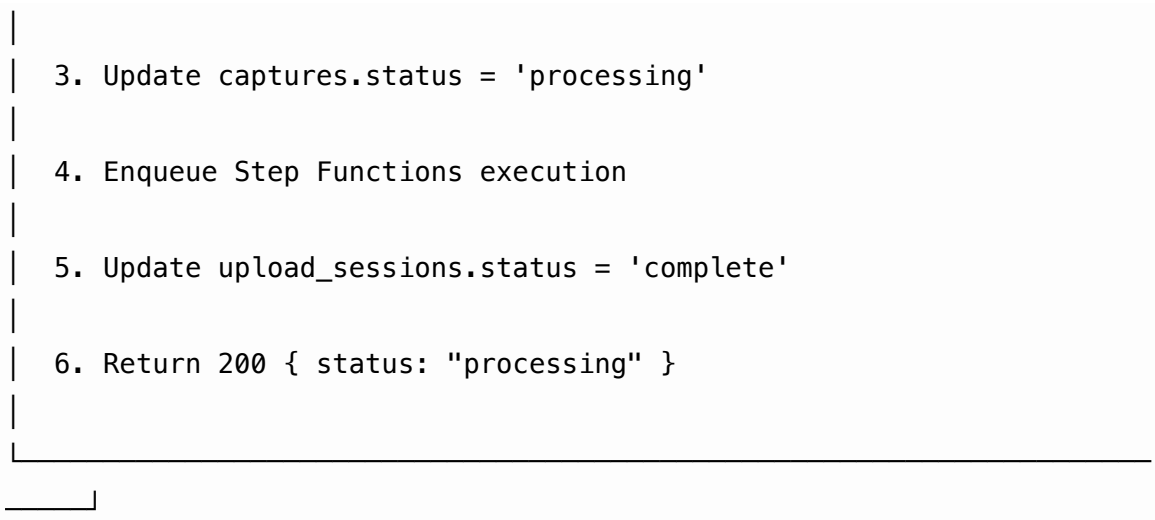


Key properties: - **Connection resilience:** mobile uses `URLSession.background` (iOS) and `WorkManager` (Android) so uploads survive airplane-mode drops - **Direct-to-S3 chunking:** the mobile app gets pre-signed S3 URLs and uploads chunks directly, bypassing the API gateway (which would add latency and cost) - **Idempotency:** `Idempotency-Key = uuid-from-session` means the mobile can retry the entire initiation on flaky network without creating duplicate captures - **Pipeline is async:** the mobile gets back immediately with an upload session; spatial AI runs in the background on GPU nodes

5.3 Capture upload — chunk-by-chunk

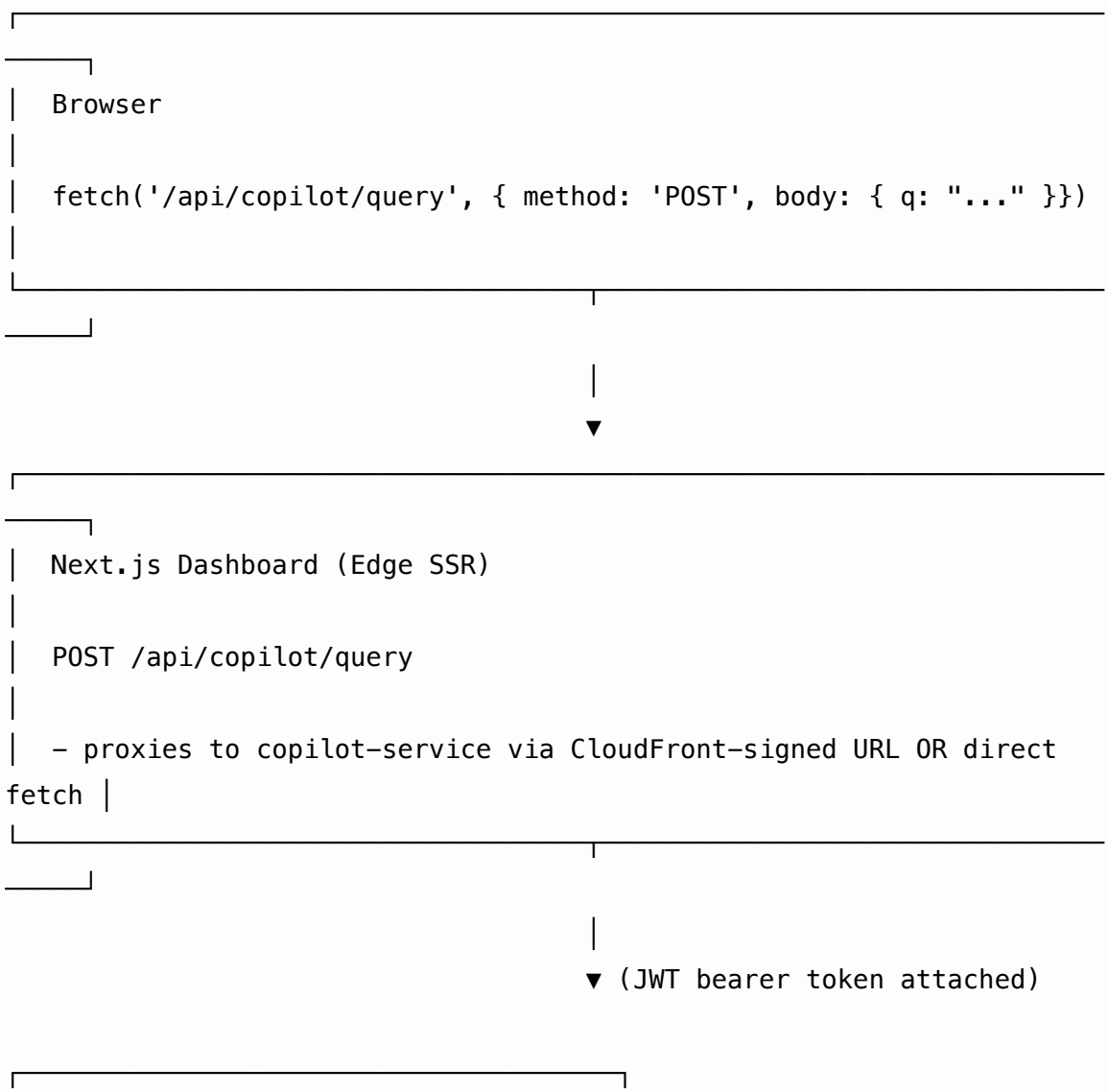
Continuing from §5.2, after the mobile gets the pre-signed URLs:

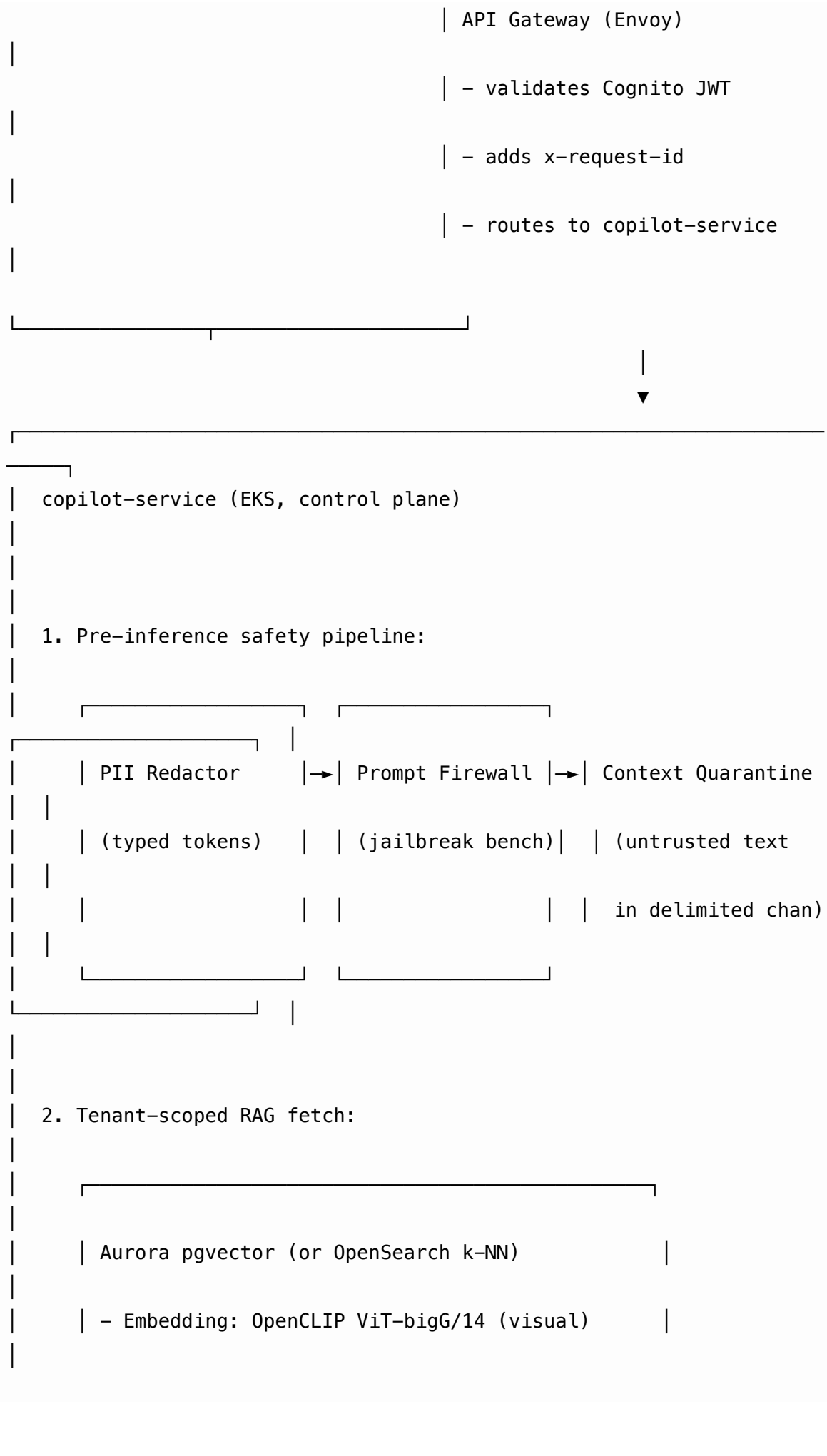




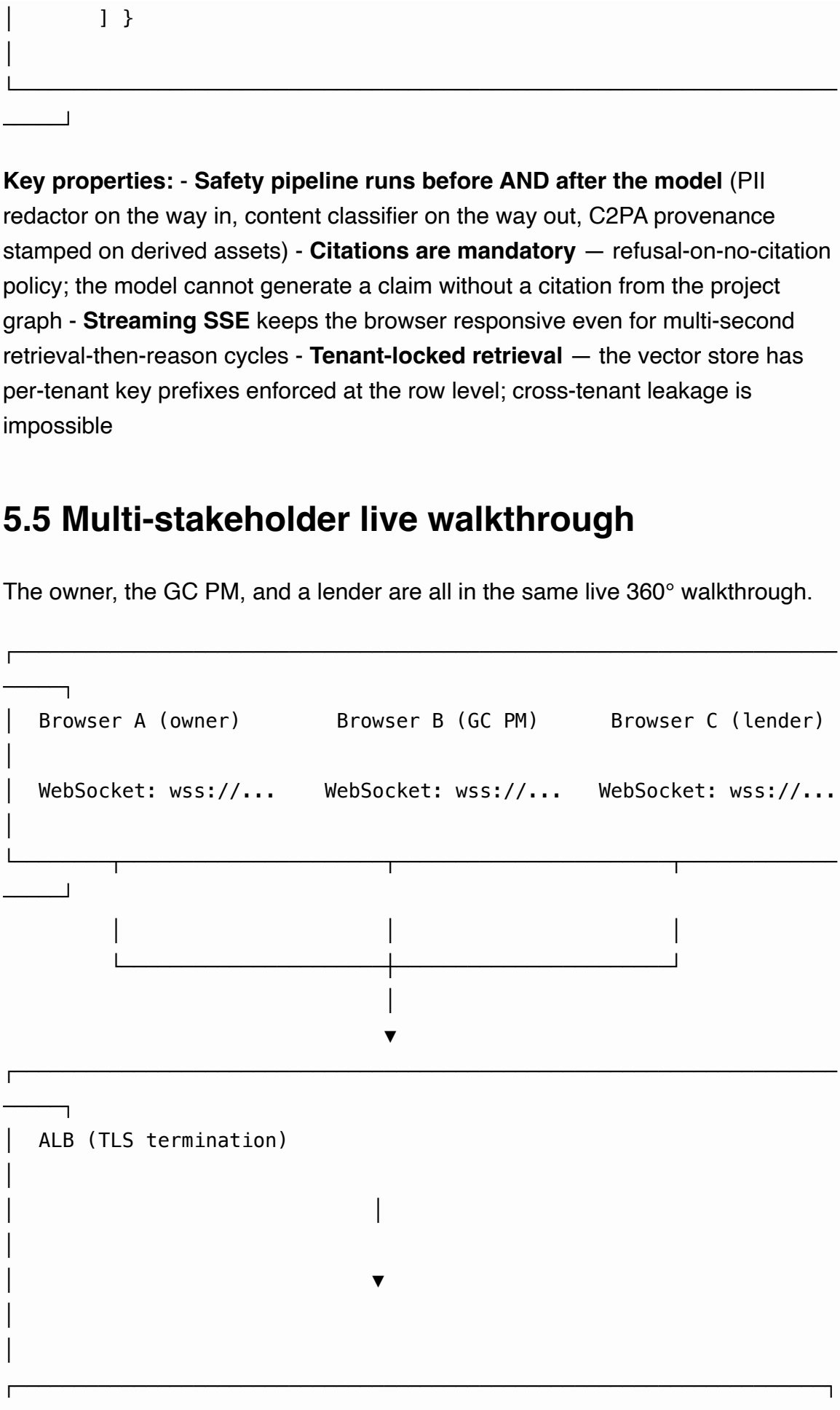
5.4 AI Copilot query

A user in the dashboard types *“show me open RFIs on Level 3 over $5k”*.




API Gateway (Envoy)
- validates Cognito JWT
- adds x-request-id
- routes to copilot-service
copilot-service (EKS, control plane)
1. Pre-inference safety pipeline:
PII Redactor
(typed tokens)
Prompt Firewall
(jailbreak bench)
Context Quarantine
(untrusted text
in delimited chan)
2. Tenant-scoped RAG fetch:
Aurora pgvector (or OpenSearch k-NN)
- Embedding: OpenCLIP ViT-bigG/14 (visual)

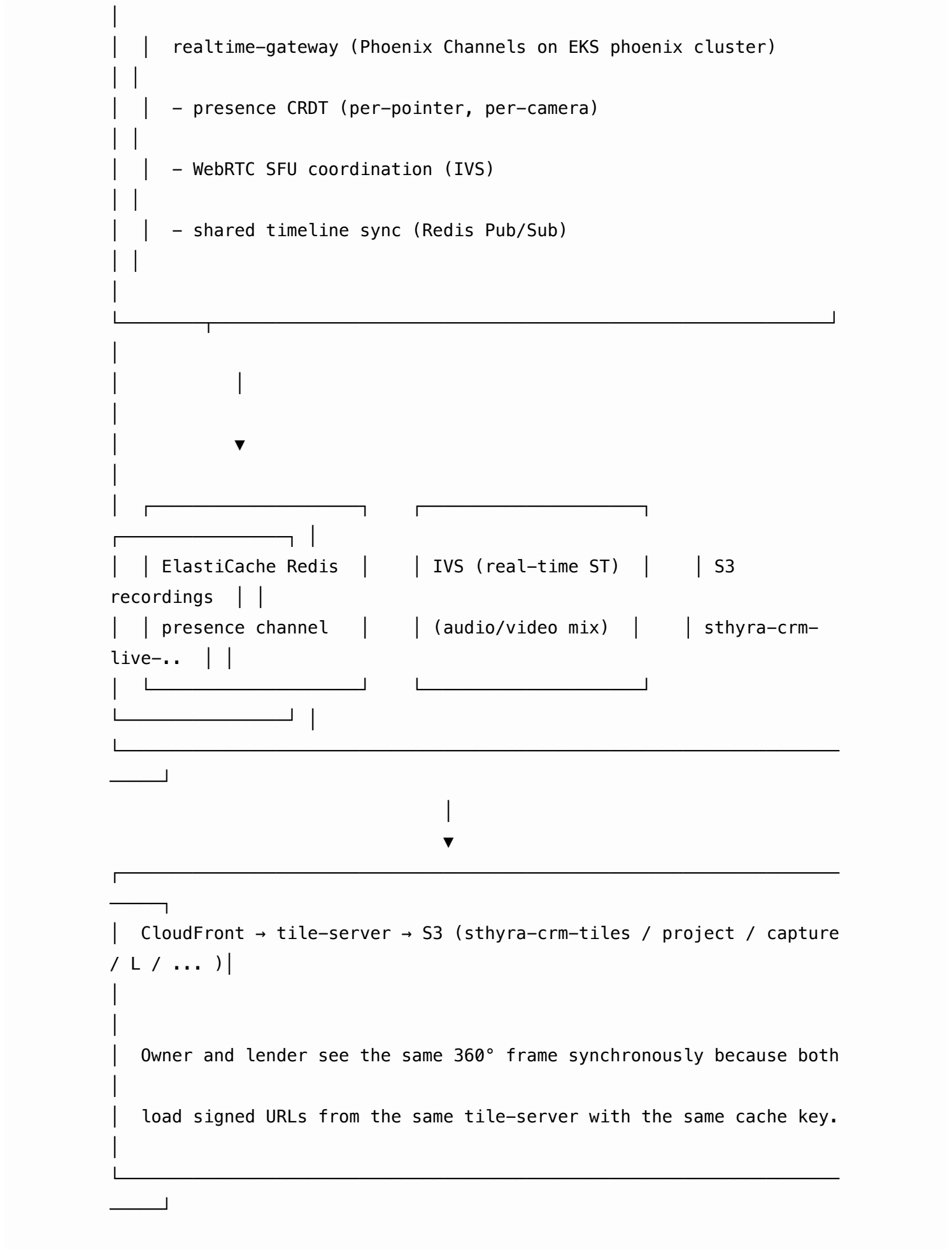
```
|     | - Embedding: BGE-M3 (text/code)                  |
|
|     | - Hybrid BM25 + dense retrieval                  |
|
|     | - Reranker: bge-reranker-v2-m3 (350M)            |
|
|     └──────────────────────────────────────────────────┘
|
|
|
|  3. LLM inference (function-calling):
|
|     ┌──────────────────────────────────────────────────┐
|
|     |  Bedrock (Claude)  OR  Self-hosted Llama 3 70B  |
|
|     |  Routing:                                        |
|
|     |   - US-GovCloud → Claude on GovCloud             |
|
|     |   - EU → Mistral on SageMaker (EU region)        |
|
|     |   - JP/KSA → Sakura-hosted Llama                 |
|
|     └──────────────────────────────────────────────────┘
|
|
|
|  4. Streaming SSE chunks → browser:
|
|     { delta: "I found 3 RFIs on Level 3 over $5k:\n",
|
|       citations: [
|
|         { type: "rfi", id: "rfi_00000007", title: "Slab
thickness...", |
|           thumbnail_url: "...",
|
|           geo: { lat: 40.7, lon: -74.0 } }
|
```



Key properties: - **Safety pipeline runs before AND after the model** (PII redactor on the way in, content classifier on the way out, C2PA provenance stamped on derived assets) - **Citations are mandatory** — refusal-on-no-citation policy; the model cannot generate a claim without a citation from the project graph - **Streaming SSE** keeps the browser responsive even for multi-second retrieval-then-reason cycles - **Tenant-locked retrieval** — the vector store has per-tenant key prefixes enforced at the row level; cross-tenant leakage is impossible

5.5 Multi-stakeholder live walkthrough

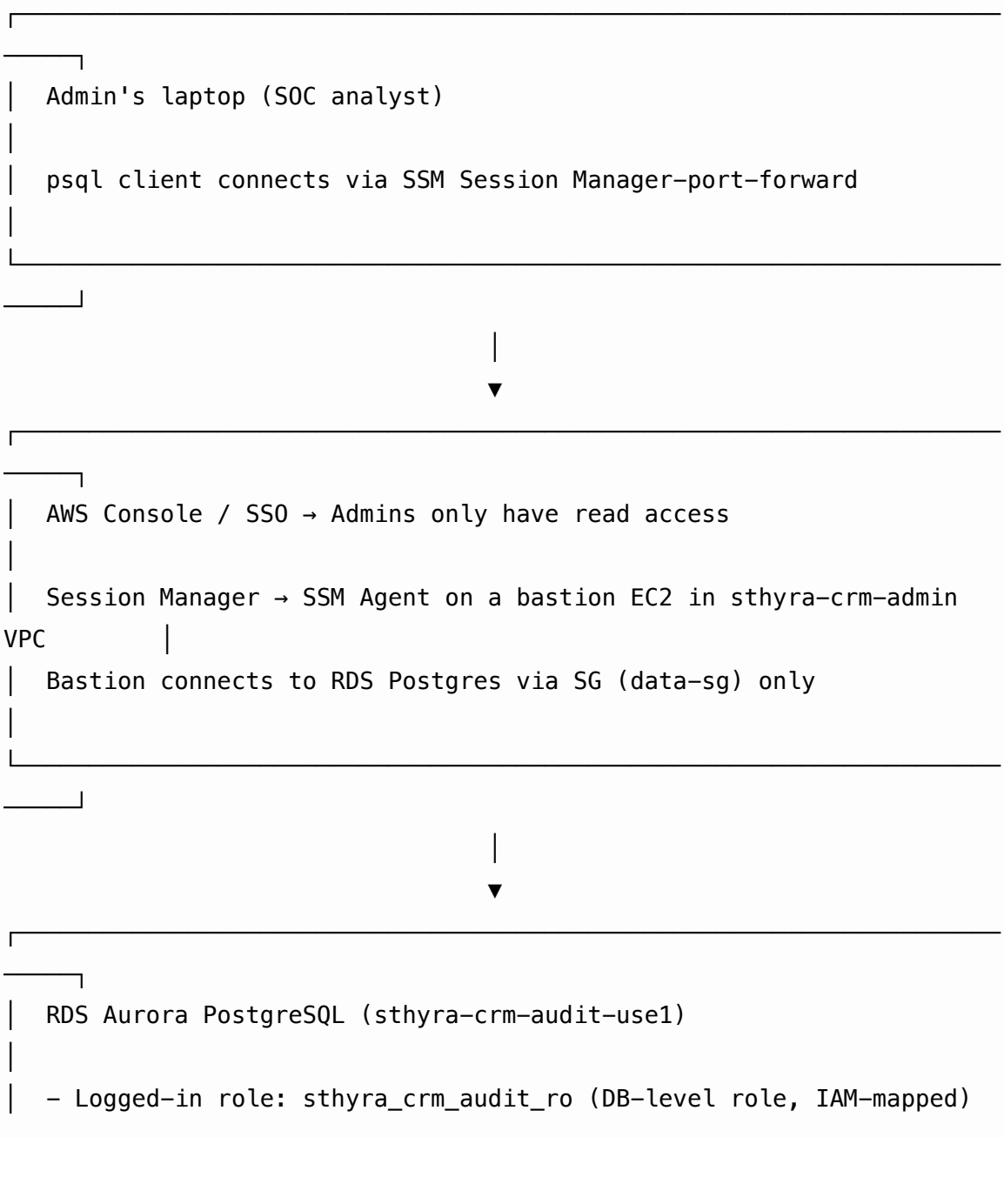
The owner, the GC PM, and a lender are all in the same live 360° walkthrough.

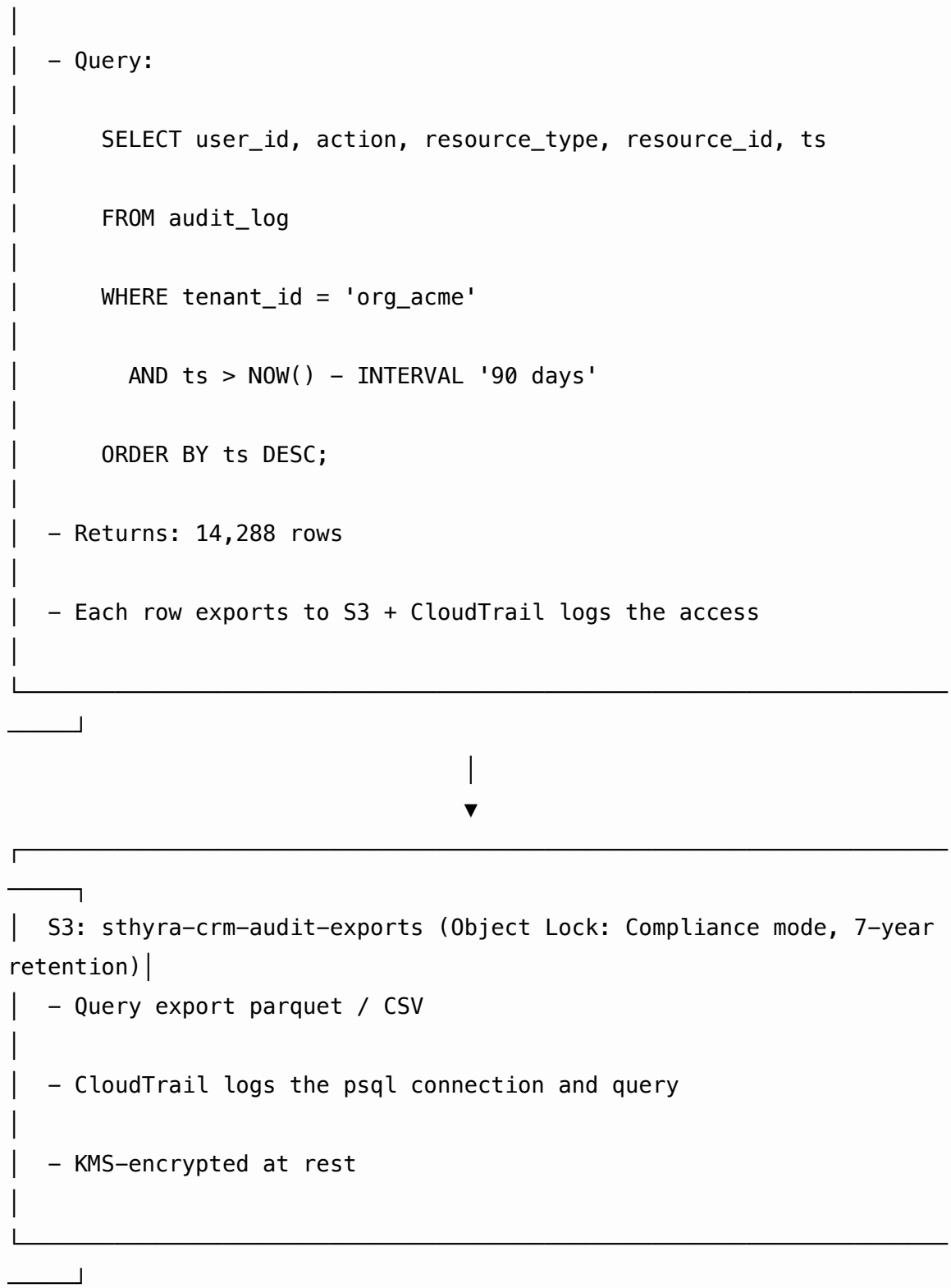

realtime-gateway (Phoenix Channels on EKS phoenix cluster)
- presence CRDT (per-pointer, per-camera)
- WebRTC SFU coordination (IVS)
- shared timeline sync (Redis Pub/Sub)
ElastiCache Redis
presence channel
IVS (real-time ST)
(audio/video mix)
S3
recordings
sthyra-crm-live-..
CloudFront → tile-server → S3 (sthyra-crm-tiles / project / capture / L / ...)
Owner and lender see the same 360° frame synchronously because both
load signed URLs from the same tile-server with the same cache key.

Key properties: - **Phoenix Channels** for soft-realtime presence (chosen for the BEAM VM's fault tolerance and natural fan-out) - **IVS (Interactive Video Service)** for real-time audio/video - **Redis Pub/Sub** as the presence bus (channel-per-walkthrough) - **S3** for recordings (`sthyra-crm-live-{region}/…`); auto-minutes via Transcribe → Bedrock summary - **Tile-server** is the same component used for non-live walkthroughs; the shared cache means owners and lenders see the same frame

5.6 Admin runs a SOC 2 audit query

The compliance team needs a query to answer: *"Show every access to a record owned by ACME Corp in the last 90 days"*.

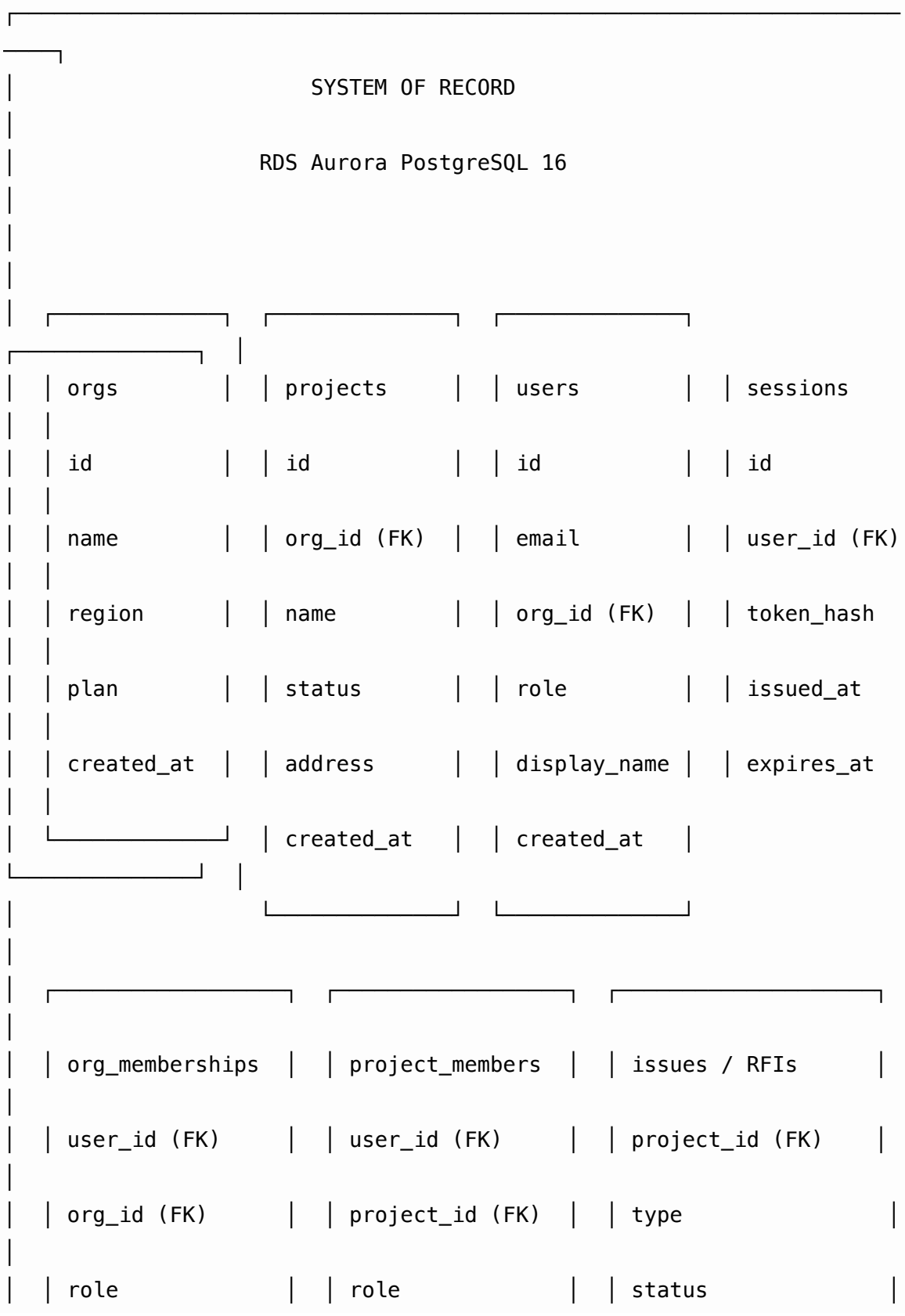


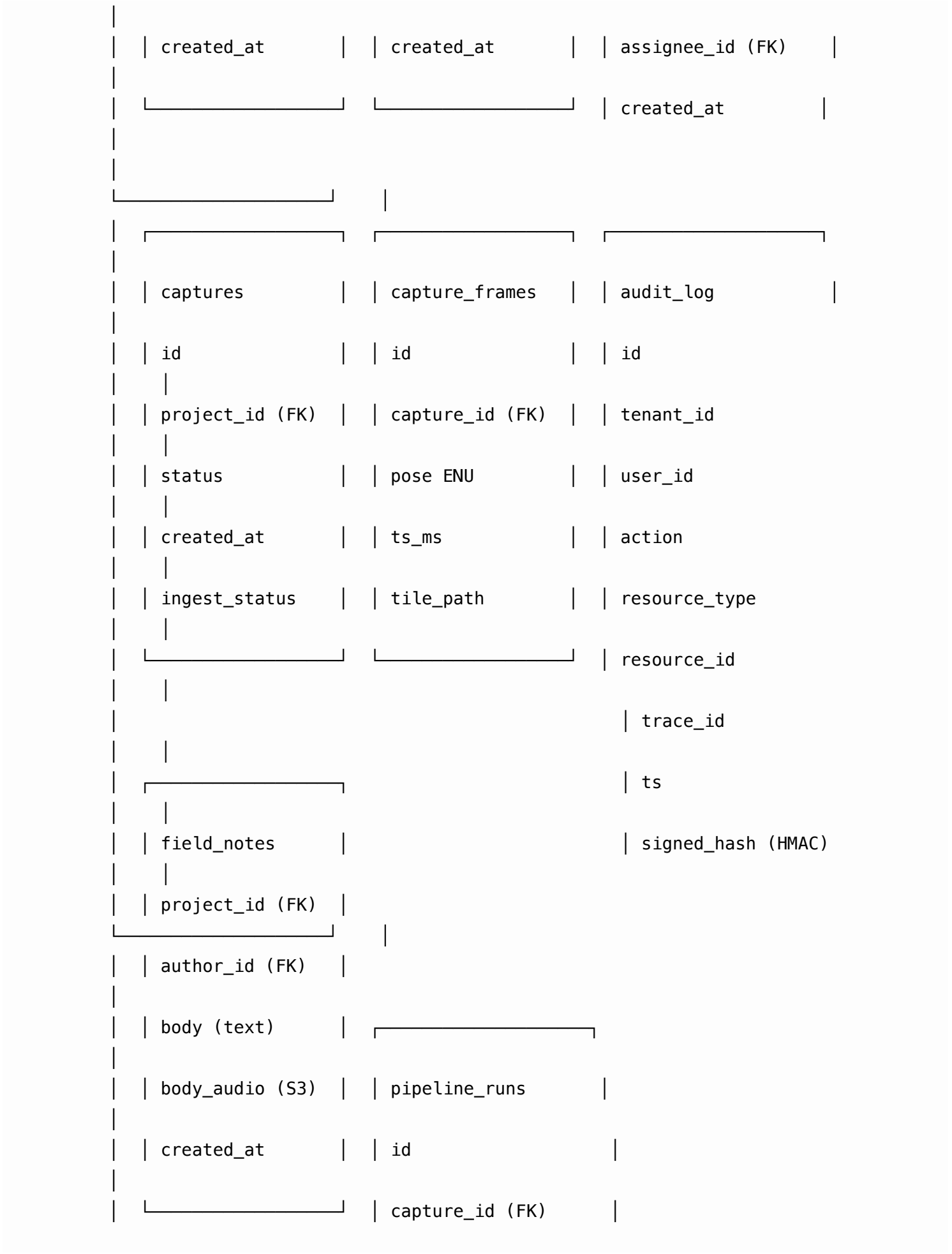


Key properties: - **No standing access.** Admins use SSM Session Manager, never long-lived credentials - **CloudTrail** records every API call (including psql queries, which generate RDS API calls) - **Object Lock + Compliance mode** prevents audit export tampering - **DB-level role separation** (`sthyra_crm_audit_ro`) prevents admins from masquerading as application users

6. Data Architecture

6.1 Logical data model




created_at
created_at
assignee_id (FK)
created_at
captures
id
project_id (FK)
status
created_at
ingest_status
capture_frames
id
capture_id (FK)
pose ENU
ts_ms
tile_path
audit_log
id
tenant_id
user_id
action
resource_type
resource_id
trace_id
ts
signed_hash (HMAC)
field_notes
project_id (FK)
author_id (FK)
body (text)
body_audio (S3)
created_at
pipeline_runs
id
capture_id (FK)

6.2 S3 bucket layout

Bucket	Purpose	Lifecycle	Encryption
`sthyra-crm-raw-360-{region}`	Raw 360° video, drone frames, video uploads	Hot 30d → IA 90d → Glacier 365d	SSE-KMS
`sthyra-crm-frames-{region}`	Extracted frames (per-capture, equirectangular + cube faces)	Hot 90d → IA 365d	SSE-KMS
`sthyra-crm-tiles-{region}`	3D tile pyramids (3D Tiles, Gaussian Splatting)	Hot 90d → IA 365d	SSE-KMS

Bucket	Purpose	Lifecycle	Encryption
`sthyra-crm-bim-{region}`	BIM models (IFC, RVT, NWD) as glTF + originals	Hot 30d → IA 365d	SSE-KMS
`sthyra-crm-attach-{region}`	User-uploaded attachments (photos, PDFs, sketches)	IA 30d → Glacier 365d	SSE-KMS
`sthyra-crm-drone-{region}`	Drone telemetry logs, orthomosaics	Hot 90d → IA 365d	SSE-KMS
`sthyra-crm-live-{region}`	Live walkthrough recordings	Hot 90d → IA 365d	SSE-KMS
`sthyra-crm-audit-exports-{region}`	Audit query exports (Object Lock Compliance, 7y)	Never (locked)	SSE-KMS + Object Lock
`sthyra-crm-log-archive-{region}`	Long-term log archive (S3 Standard-IA after 30d)	Hot 30d → IA 365d	SSE-KMS
`sthyra-crm-customer-exports-{region}`	Per-tenant export buckets (BYO data)	Customer-defined	Customer CMK

All buckets are private. Access via CloudFront signed URLs or S3 access points only. Cross-region replication active for `sthyra-crm-raw-360-` and `sthyra-crm-audit-exports-` (encryption-at-rest is preserved across regions).

6.3 TimescaleDB (capture telemetry)

```
┌──────────────────────────────────────────────┐
│  TimescaleDB (ext. on RDS Aurora)            │
│                                              │
│  hypertable: capture_telemetry               │
│    ts           timestamptz   (partition key) │
```

```
│    capture_id   text                                │
│    user_id      text                                │
│    event_type   text  (gps, imu, ar_drop, ...)      │
│    payload      jsonb                               │
│  ──────────────────────────────────────────────────│
│  hypertable: device_pings                           │
│    ts           timestamptz                         │
│    user_id      text                                │
│    battery_pct  smallint                            │
│    thermal_state text                               │
│  ──────────────────────────────────────────────────│
│  hypertable: sync_events                            │
│    ts           timestamptz                         │
│    device_id    text                                │
│    user_id      text                                │
│    event        text                                │
│    bytes_synced bigint                              │
└─────────────────────────────────────────────────────┘
```

Retention: 90 days hot, 2 years compressed (continuous aggregates).

6.4 ClickHouse (analytics)

```
┌─────────────────────────────────────────────────────┐
│  ClickHouse Cloud (or Redshift Serverless)          │
│                                                     │
│  events                                             │
│    ts              DateTime                         │
│    tenant_id       String                           │
│    user_id         String                           │
│    event_type      LowCardinality(String)           │
│    properties      String (JSON)                    │
│  ──────────────────────────────────────────────────│
│  rum (real user metrics — frontend telemetry)       │
│    ts              DateTime                         │
│    session_id      String                           │
│    route           String                           │
│    lcp_ms          UInt32                           │
│    fid_ms          UInt32                           │
│    cls             Float32                          │
```

```
│  ──────────────────────────────────────────────┤
│  cost_rollup                                   │
│    tenant_id       String                      │
│    service         String                      │
│    day             Date                        │
│    ai_minutes      Float64                     │
│    egress_gb       Float64                     │
│    storage_gb      Float64                     │
└────────────────────────────────────────────────┘
```

6.5 Redis usage

Cluster	Engine	Purpose	Eviction	TTL
`sthyra-crm-session-{region}`	Redis 7	Session store, idempotency cache, idempotency keys	allkeys-lru	1h sessions, 24h idem
`sthyra-crm-presence-{region}`	Redis 7	Phoenix Channels presence pub/sub	volatile-lru	ephemeral
`sthyra-crm-rate-{region}`	Redis 7	Token-bucket rate limiting per `(userId, route)`	volatile-lru	1m
`sthyra-crm-flow-{region}`	Redis 7	Idempotency and dedup keys for upstream integrations	volatile-lru	24h

Cluster mode: 6 shards × 2 replicas per primary. AOF persistence on `sthyra-crm-session-` and `sthyra-crm-flow-` only.

7. Compute Architecture

7.1 EKS cluster topology

Cluster	Roles	Node groups	Region deployment
`sthyra-crm-control-use1`	All stateless services (Phase 0 + Phase 1 control)	system, app, gpu	us-east-1
`sthyra-crm-realtime-use1`	Phoenix Channels realtime gateway	realtime	us-east-1
`sthyra-crm-gpu-use1`	imgproc-service workers (P5/G5)	gpu	us-east-1
`sthyra-crm-batch-use1`	Step Functions, ETL, scheduled jobs	batch	us-east-1

Mirrored in `eu-west-1`, `ap-southeast-2`, `ap-northeast-1`, `ksa-central-1`.

7.2 Node groups

Group	AMI	Capacity	Disk	Use
`system`	Bottlerocket EKS-optimized	t4g.medium × 3	20 GB gp3	Karpenter, CoreDNS, Ingress-NGINX
`app`	Bottlerocket EKS-optimized	m6i.xlarge × 3 (min), 30 (max)	100 GB gp3	Stateless services
`app-arm`	Bottlerocket (arm64)	m6g.xlarge × 3 (min), 20 (max)	100 GB gp3	Cost-optimized for low-CPU services
`gpu-a10g`	DLAMI	g5.xlarge × 1 (min), 8 (max)	500 GB gp3	SAM-2 inference (online)

Group	AMI	Capacity	Disk	Use
`gpu-a100`	DLAMI	p4d.24xlarge × 0 (min), 4 (max)	1 TB gp3	3DGS training (batch)
`gpu-h100`	DLAMI	p5.48xlarge × 0 (min), 2 (max)	2 TB gp3	Heavy batch (LLM fine-tuning, model upgrades)
`realtime`	Bottlerocket EKS-optimized	c6i.2xlarge × 3 (min), 12 (max)	50 GB gp3	Phoenix Channels cluster
`batch`	Bottlerocket EKS-optimized	m6i.2xlarge × 0 (min), 10 (max)	100 GB gp3	Step Functions, ETL

Karpenter manages the `app`, `app-arm`, `gpu-a10g`, `gpu-a100`, `gpu-h100`, and `batch` node groups. **Cluster Autoscaler** manages `system` and `realtime` (predictable baseline).

7.3 Deploy topology per service

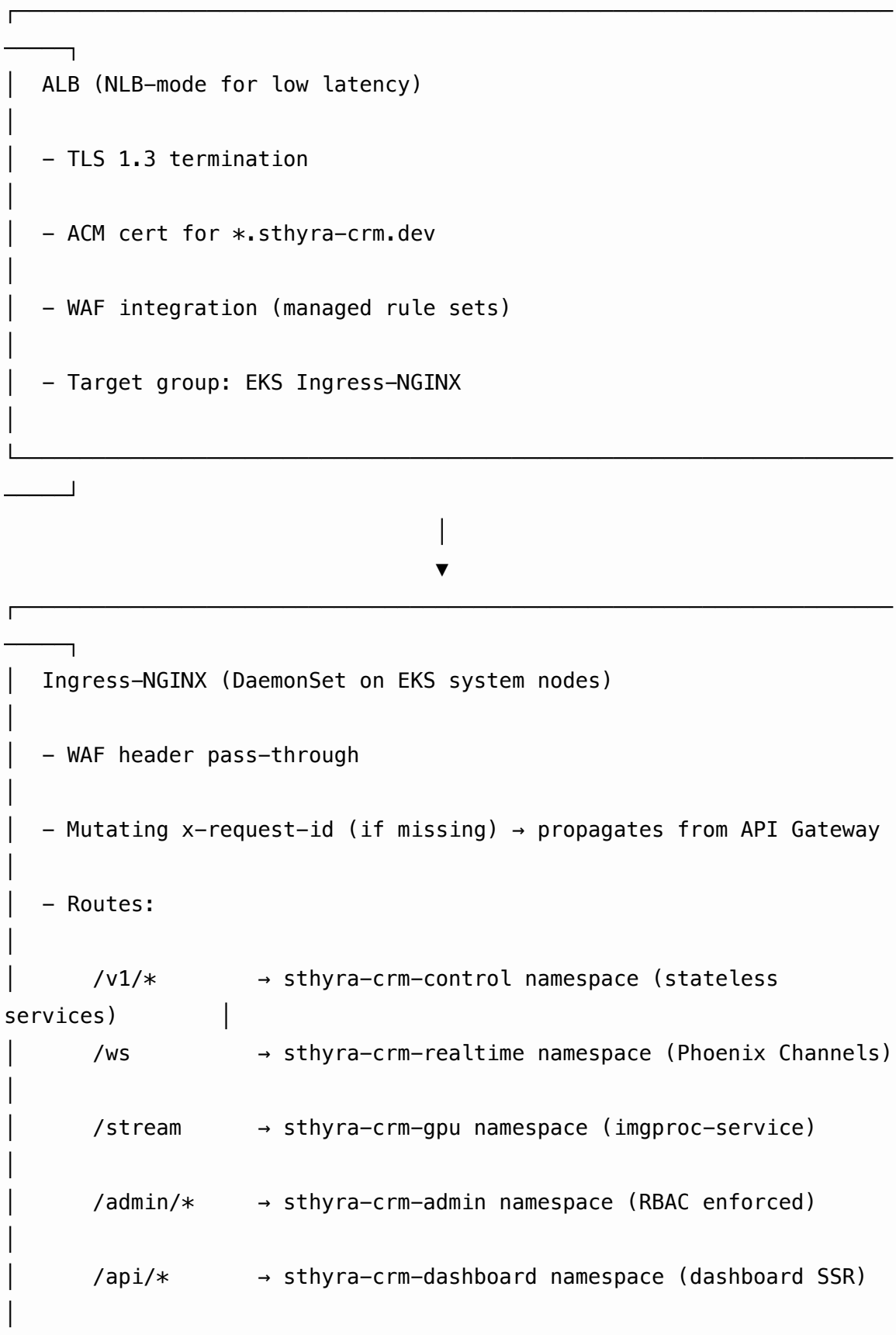
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: org-service
  namespace: sthyra-crm
spec:
  replicas: 3
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 0
  template:
    spec:
      serviceAccountName: org-service
      topologySpreadConstraints:
        - maxSkew: 1
          topologyKey: topology.kubernetes.io/zone
```

```
      whenUnsatisfiable: DoNotSchedule
  containers:
    - name: org-service
      image: <account>.dkr.ecr.us-east-1.amazonaws.com/org-
    service:v0.1.0
      resources:
        requests: { cpu: 100m, memory: 256Mi }
        limits:   { cpu: 500m, memory: 512Mi }
      readinessProbe:
        httpGet: { path: /v1/health, port: 8080 }
        initialDelaySeconds: 5
        periodSeconds: 5
      livenessProbe:
        httpGet: { path: /v1/health, port: 8080 }
        initialDelaySeconds: 30
        periodSeconds: 30
      env:
        - name: DATABASE_URL
          valueFrom:
            secretKeyRef:
              name: org-service-secrets
              key: database-url
        - name: SERVICE_NAME
          value: org-service
      securityContext:
        runAsNonRoot: true
        runAsUser: 10001
        readOnlyRootFilesystem: true
        allowPrivilegeEscalation: false
        capabilities:
          drop: [ALL]
```

Key properties: - **Zero-trust pod:** non-root, read-only root FS, all capabilities dropped - **IRSA (IAM Roles for Service Accounts):** the `org-service` ServiceAccount is annotated to assume an IAM role via OIDC, so the pod can use `aws-sdk` without static credentials - **Topology spread:** even distribution across AZs - **Rolling updates** with `maxUnavailable: 0` = zero-downtime deploys

7.4 Ingress

API Gateway → ALB (TLS 1.3) → Ingress-NGINX → Service.



```
│  - Rate limiting via NGINX annotations (defense-in-depth)
│
└──────────────────────────────────────────────────────────────────────────────
──────┘
```

7.5 Step Functions (capture pipeline)

```
┌──────────────────────────────────────────────────────────────────────────────
──────┐
│  Step Functions state machine: CapturePipeline
│
│
│
│  StartAt: DecodeFrame
│
│  DecodeFrame → ExtractFrames (Lambda)
│
│  ExtractFrames → SuperPointFeatures (Lambda + GPU)
│
│  SuperPointFeatures → SfM_GLOMAP (ECS RunTask)
│
│  SfM_GLOMAP → MVS_OpenMVS (ECS RunTask)
│
│  MVS_OpenMVS → MeshPoisson (ECS RunTask)
│
│  MeshPoisson → ThreeDGS (ECS RunTask)
│
│  ThreeDGS → SegmentSAM2 (ECS RunTask)
│
│  SegmentSAM2 → AlignToBIM (ECS RunTask)
│
│  AlignToBIM → Publish (Lambda)
│
│  Publish → SNS → capture-service (update DB + emit WebSocket)
│
│
│
│  Failure handling:
│
│  - Any state error → Catch → DLQ (SQS) → on-call page
```

```
│
│  - SfM/MVS retries with smaller image pool after 3 failures
│
│  - User visible state: 'processing' → 'succeeded' or 'failed'
│
└──────────────────────────────────────────────────────────────────────────────
─────┘
```

8. Networking & Service Mesh

8.1 Service mesh: App Mesh vs Istio (decision)

Decision: App Mesh with Envoy sidecars, *not* Istio.

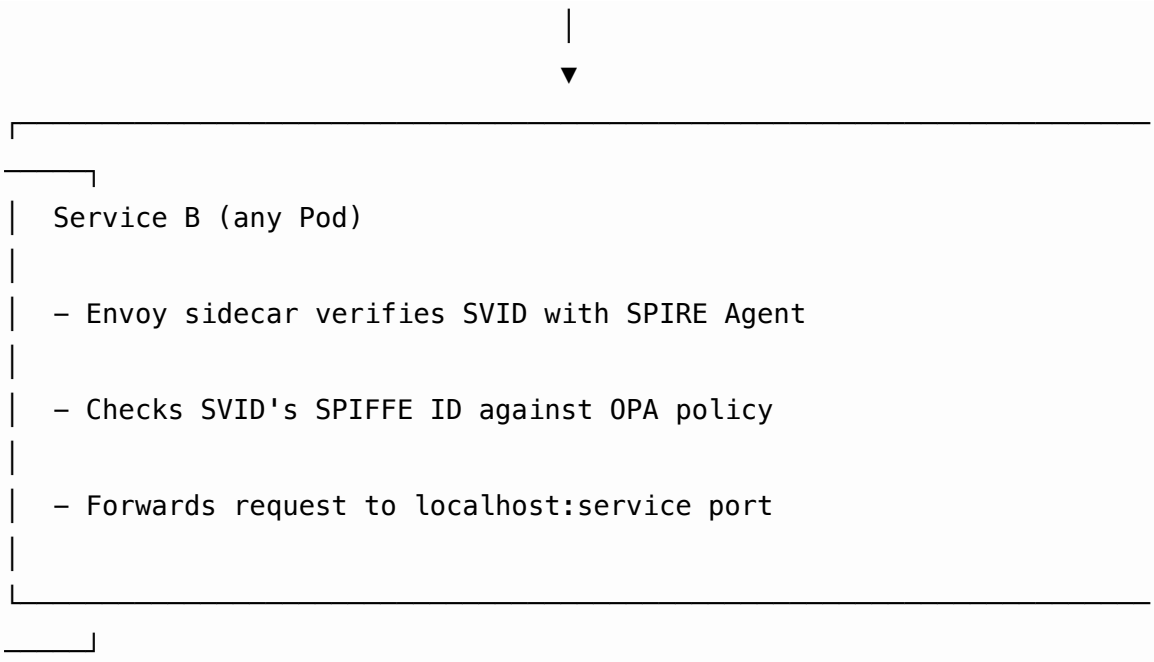
Reasons: - App Mesh is **AWS-native** (no third-party control plane to operate) - **Lower operational cost** (no Istio control plane, no extra RBAC) - **Native integration** with X-Ray, CloudWatch, IAM - **AWS-supported** — backed by AWS engineering - Master plan §10 specifies “service mesh only where operationally justified (not added by default)”

For Phase 0 and Phase 1, we don’t deploy a mesh. East-west traffic is HTTP over the cluster-internal network. App Mesh becomes relevant at Phase 2 when mTLS + SPIFFE is mandated.

8.2 Service-to-service auth (Phase 2+)

When mTLS is mandated:

```
┌──────────────────────────────────────────────────────────────────────────────
──────┐
│  Service A (any Pod)
│
│  - Envoy sidecar injects SPIFFE SVID (TTL ≤ 1h)
│
│  - Outbound request: HTTP + X-SPIFFE-Token: <SVID>
│
└──────────────────────────────────────────────────────────────────────────────
─────┘
```



Deployment: SPIRE Server runs as a StatefulSet; SPIRE Agent runs as a DaemonSet; each service has a SPIRE Registration Entry. The `@sthyra-crm/auth` package already implements the verify side (§4.6 of STHYRA-PHASE-0-REPORT.md); the SPIRE integration is a Phase 2 wiring task.

8.3 Inter-AZ traffic

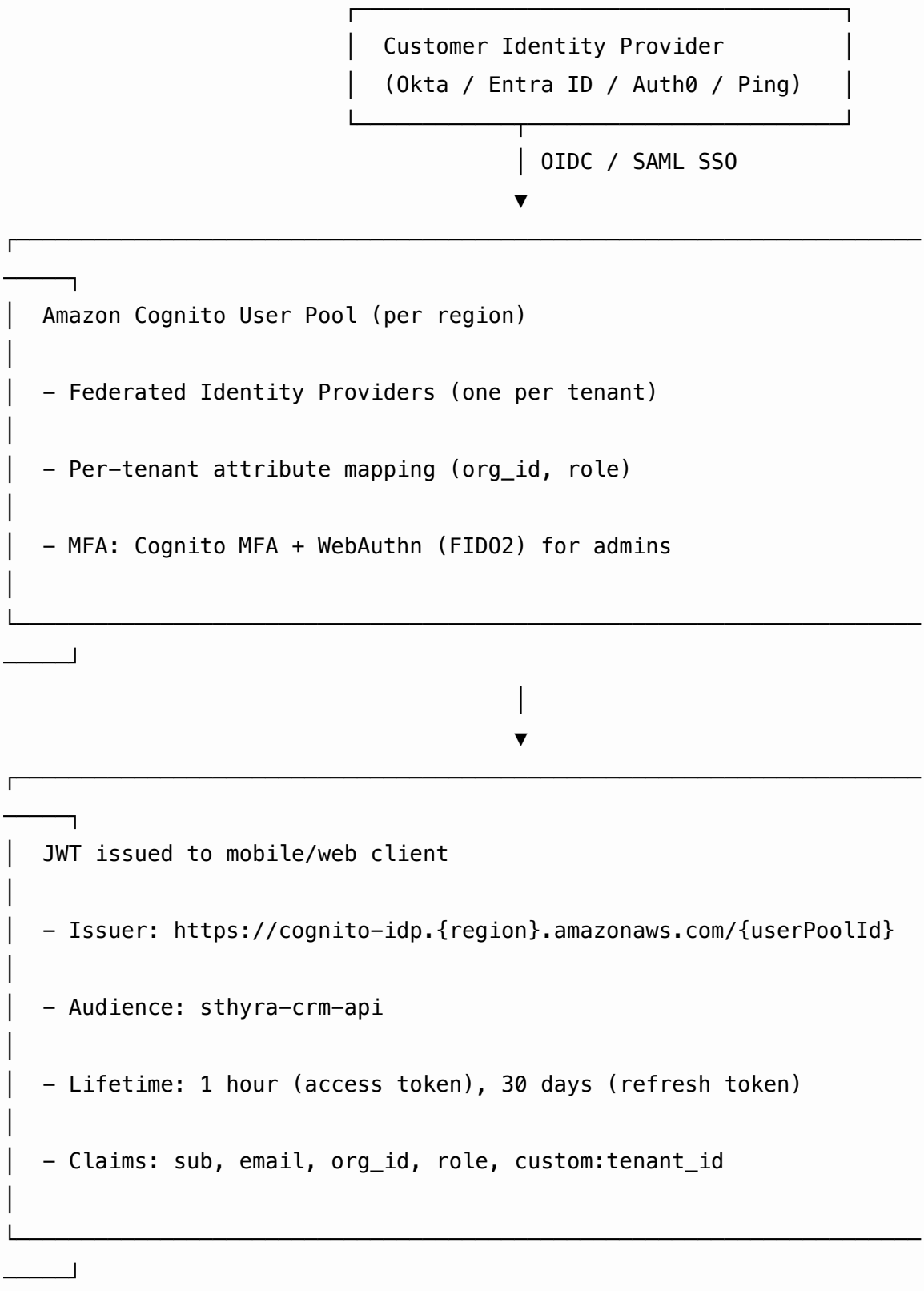
- All services run in **3 AZs** for HA
- Aurora PostgreSQL uses **Multi-AZ** with sync replicas
- ElastiCache Redis uses **Multi-AZ** with automatic failover
- S3 is **regional** (multi-AZ transparent)
- OpenSearch uses **Multi-AZ** with 3 dedicated master nodes

8.4 Inter-region traffic

- **S3 Cross-Region Replication** for `sthyra-crm-raw-360-` and `sthyra-crm-audit-exports-`
- **Aurora Global Database** for cross-region read replicas (RPO ≤ 1s)
- **Route 53 latency-based routing** across regions
- **Transit Gateway peering** between commercial regions (us-east-1 ↔ us-west-2 ↔ eu-west-1 ↔ ap-southeast-2 ↔ ap-northeast-1)
- **GovCloud (us-gov-east-1)** is isolated — no transit peering to commercial

9. Identity & Access

9.1 User identity (Cognito + external IdP)



Verification flow: 1. User logs in via their IdP (Okta / Entra) 2. IdP returns code to mobile/web 3. Mobile/web exchanges code for Cognito JWT 4. Mobile/web sends JWT as `Authorization: Bearer <jwt>` to API Gateway 5. API Gateway validates JWT signature, claims, expiry 6. Cognito user pool lookup populates `req.principal`

9.2 Workload identity (IAM Roles for Service Accounts)

Every EKS ServiceAccount is annotated with an IAM role. This is the **AWS-native workload identity** mechanism; SPIFFE is the **cross-cloud** alternative.

```
apiVersion: v1
kind: ServiceAccount
metadata:
  name: org-service
  namespace: sthyra-crm
  annotations:
    eks.amazonaws.com/role-arn: arn:aws:iam::123456789012:role/sthyra-
        crm-org-service
```

The IAM role has a trust policy that only allows the `org-service` SA to assume it (via OIDC issuer `oidc.eks.us-east-1.amazonaws.com/...`). It has permissions scoped to exactly what the service needs:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": ["secretsmanager:GetSecretValue"],
      "Resource": "arn:aws:secretsmanager:us-east-
        1:123456789012:secret:org-service-*"
    },
    {
      "Effect": "Allow",
      "Action": ["rds-db:connect"],
      "Resource": "arn:aws:rds:us-east-1:123456789012:db:sthyra-crm-
        control"
    }
  ]
}
```

9.3 Tenant isolation

Tenant isolation is enforced at **five layers**:

1. **Database:** every row has `tenant_id`; row-level security policies filter rows by `current_setting('app.tenant_id')`
2. **Redis:** keys are prefixed with `tenant_id`; ACLs disallow cross-tenant access
3. **S3:** objects have `tenant_id` in the key; access points have per-tenant policies
4. **KMS:** every tenant has a Customer Managed Key (CMK); per-object encryption uses the tenant's CMK
5. **API Gateway:** JWT has `tenant_id` claim; every service routes it through to DB queries

10. Observability

10.1 The three pillars

```
┌──────────────────────────────────────────────────────────────────────────────
─────┐
│                            METRICS
│
│  - CloudWatch Metrics (AWS-native)
│
│  - Prometheus + Grafana (SLO dashboards)
│
│  - 1s resolution for control-plane alerts
│
│  - Custom metrics: SLO burn rate, error budget, queue depth
│
└──────────────────────────────────────────────────────────────────────────────
─────┘

┌──────────────────────────────────────────────────────────────────────────────
─────┐
│                            LOGS
│
│  - Structured JSON via @sthyra-crm/observability
```

```
│
│  - CloudWatch Logs (control plane)
│
│  - Kinesis Firehose → S3 (long-term archive)
│
│  - OpenSearch (full-text search, ad-hoc)
│
│  - Every log line carries request_id, service, ts, level, msg
│
└──────────────────────────────────────────────────────────────────────────────┘


┌──────────────────────────────────────────────────────────────────────────────┐
│                                TRACES
│
│  - OpenTelemetry SDK in every service
│
│  - AWS X-Ray collector (DaemonSet)
│
│  - 2.5% sampling default, 100% sampling for errors
│
│  - Trace context propagated via W3C `traceparent` header
│
└──────────────────────────────────────────────────────────────────────────────┘
```

10.2 SLO definitions

Service	SLI	SLO	Error budget
All HTTP APIs	p99 latency	< 500ms	0.1% per month
All HTTP APIs	availability	99.9%	43m/month
Capture ingest	chunk upload success	99.5%	0.5% per month
Spatial AI pipeline	capture-to-publish latency	p50 < 1.3h, p95 < 3h	n/a (latency-bound)
Copilot	query latency (p95)	< 4s	n/a

Service	SLI	SLO	Error budget
Realtime gateway	WebSocket connection stability	99.5%	0.5% per month

10.3 Alert routing

```
SEV1 (production down, data loss, security breach) → PagerDuty → on-call SRE
  → 5 min ack, 30 min mitigate
SEV2 (major feature broken, SLO burn > 2x) → PagerDuty → on-call engineer
  → 15 min ack, 4h mitigate
SEV3 (minor broken, SLO burn > 1x) → Slack #oncall, no page
  → next business day
SEV4 (cosmetic, observation) → Linear ticket, no notification
```

10.4 Cost observability

Per-tenant cost attribution via Cost and Usage Reports (CUR) + Athena:

```sql
-- per-tenant, per-service, per-day cost
SELECT
  tenant_id,
  line_item_product_code,
  DATE(line_item_usage_start_date) AS day,
  SUM(line_item_unblended_cost) AS cost_usd
FROM sthyra_crm_cur
WHERE line_item_usage_start_date >= NOW() - INTERVAL '30 days'
GROUP BY 1, 2, 3
ORDER BY day DESC;
```

Per-tenant dashboard (Grafana) shows: - AI-minutes used (from service-level metrics) - Storage GB (from S3 inventory) - Egress GB (from CloudFront logs) - Total cost (from CUR)

11. Security & Compliance

11.1 Layered defenses

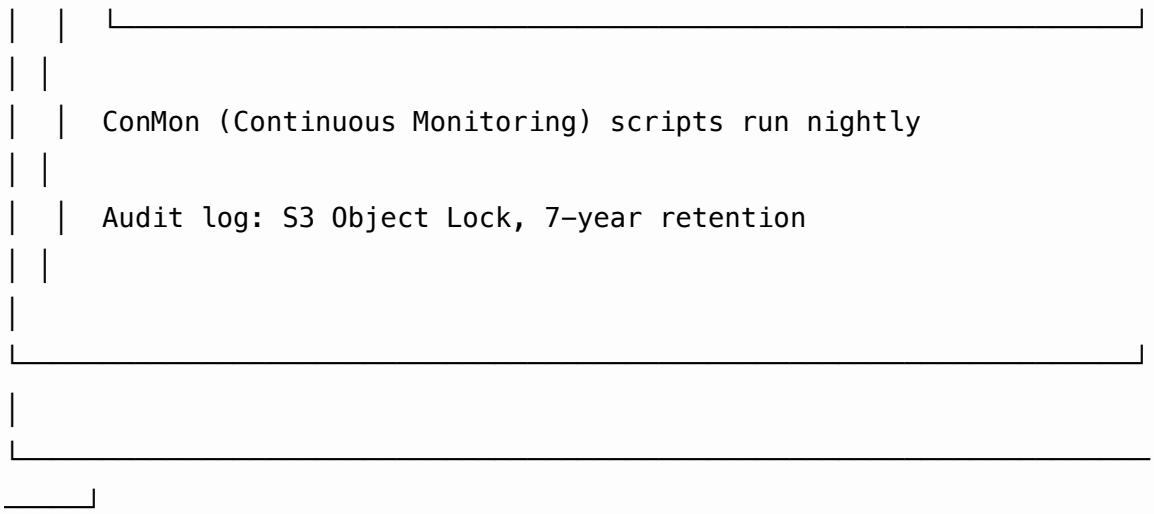
Layer	Defense	AWS Service
Edge	WAF managed rule sets, rate limiting, geo-restriction	AWS WAF, Shield Advanced
Transport	TLS 1.3 everywhere, ACM cert rotation	ACM, CloudFront
Authentication	Cognito + OIDC + SAML SSO + MFA	Cognito
Authorization	RBAC + ABAC + OPA policies	OPA (in EKS)
Network	VPC isolation, security groups, VPC endpoints	VPC, PrivateLink
Workload	SPIFFE/SVID mTLS (Phase 2+)	SPIRE
Data at rest	SSE-KMS with per-tenant CMK	KMS, CloudHSM (root)
Data in transit	TLS 1.3, mTLS (Phase 2+)	ACM, SPIRE
Secrets	AWS Secrets Manager, auto-rotation 90d	Secrets Manager
Application	SAST, DAST, SCA, dep scanning, SBOM	GitHub Actions, SonarQube
Audit	CloudTrail, S3 Object Lock Compliance	CloudTrail, S3
Penetration testing	Quarterly third-party	(external vendor)

11.2 Compliance posture

Compliance	Status	Target	AWS services used
SOC 2 Type II	In progress (Phase 1)	Type II report Q2 2027	CloudTrail, Config, IAM

Compliance	Status	Target	AWS services used
ISO 27001	In progress (Phase 2)	Q4 2027	(same as SOC 2)
FedRAMP Moderate	In progress (Phase 2)	Authorization Q4 2027	GovCloud, CloudHSM, ConMon
HIPAA	In progress (Phase 3)	Q2 2028	(same as SOC 2)
GDPR	Compliance-ready	Continuous	EU region, data residency, DSR workflows
CCPA/CPRA	Compliance-ready	Continuous	(same as GDPR)

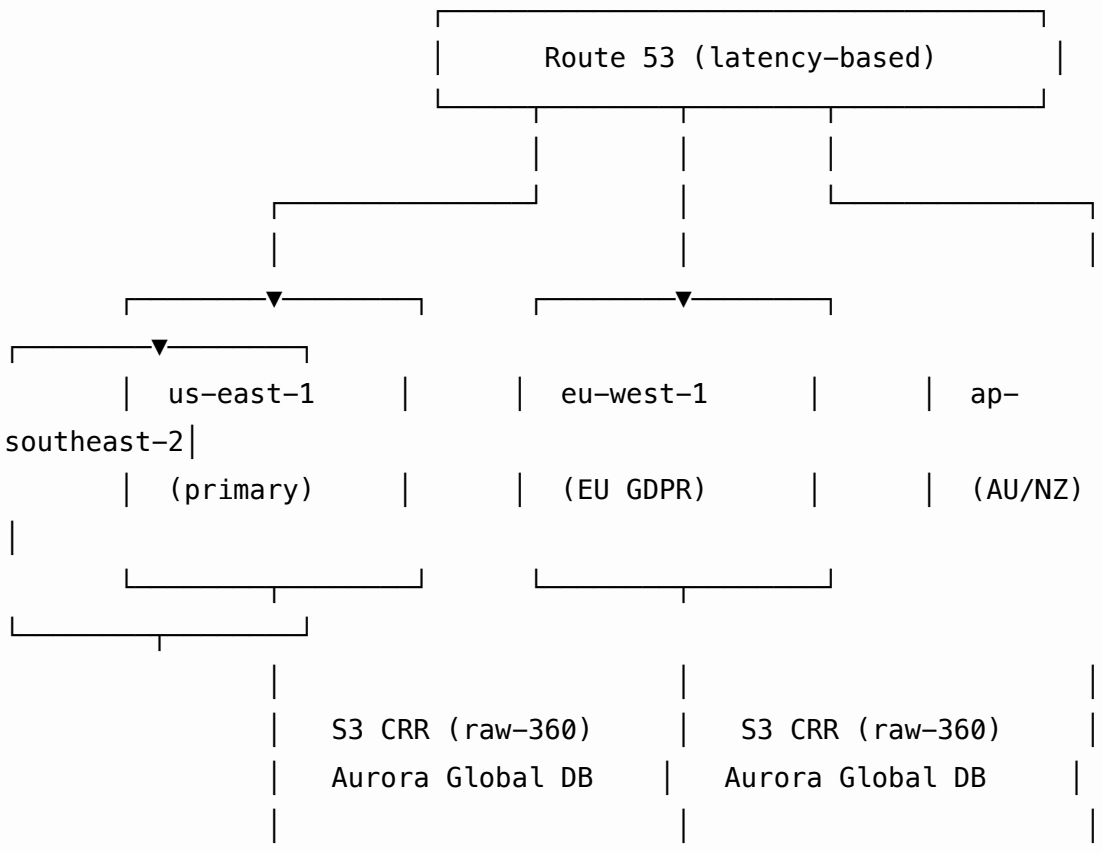
11.3 FedRAMP boundary (us-gov-east-1)

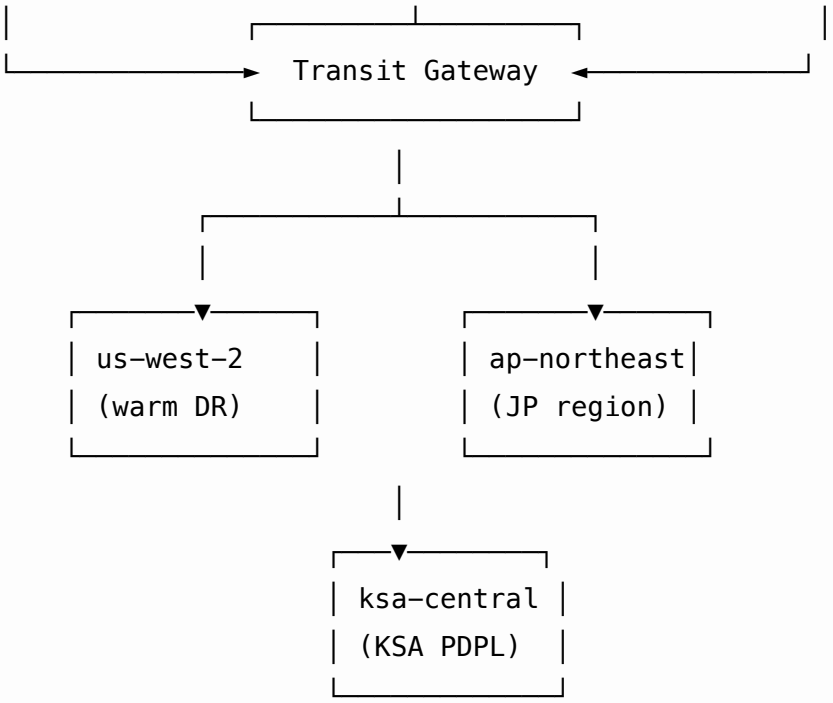


FedRAMP authorization requires: - 12–18 months of evidence collection - ConMon team hired before GA - Annual penetration test by FedRAMP-accredited 3PAO - POA&M (Plan of Action & Milestones) tracking

12. Multi-Region & Disaster Recovery

12.1 Region topology





12.2 RPO / RTO targets

Service tier	RPO	RTO	Strategy
Tier 1 (control plane APIs)	15 min	1 hour	Multi-AZ + Aurora Global DB
Tier 2 (capture pipeline)	1 hour	4 hours	S3 CRR + Step Functions replay
Tier 3 (analytics, audit logs)	4 hours	24 hours	S3 IA + Glacier + lazy replication

12.3 Active-active vs active-passive

Layer	Strategy
API Gateway	Active-active (Route 53 latency-based)
Aurora Postgres	Active-active for reads (Global DB), active-passive for writes (single primary per region)
S3	Active-active (CRR)

Layer	Strategy
Redis	Active-passive per region (Cluster Mode with replicas)
GPU pipeline	Active-passive (only one region runs the pipeline at a time, others stand by)

12.4 DR drill cadence

- Quarterly: full-region failover drill (simulated by Route 53 health check failure or maintenance page)
- Monthly: Aurora Global DB promote read replica to primary
- Weekly: S3 cross-region replication lag check
- Daily: automated backup integrity check

13. Cost Model

Per-tenant unit economics on AWS, scaled to 1,000 active customers.

13.1 Variable cost per capture

Component	Service	Cost per capture
Storage (raw 360, 5 GB avg)	S3 Standard → IA after 30d	$0.012/GB × 5 GB = $0.060
Compute (decode + SfM + MVS + 3DGS)	Lambda + EC2 GPU spot	$0.65 / capture (12 min p5.4xlarge avg)
Embeddings (BGE-M3 + OpenCLIP)	Bedrock Titan	$0.002 / capture
Inference (Copilot when used)	Bedrock Claude	$0.14 / active user / day
CDN egress	CloudFront	$0.05 / GB × 2 GB avg = $0.10
Step Functions	Step Functions	$0.025 per state transition
Total per capture		**~$0.95**

13.2 Per-tenant monthly cost (Pro tier)

Component	Service	Cost per tenant/month
Seat (5 users avg)	Internal allocation	$25 / user / month × 5 = $125
100 captures / month	Per-capture cost	$0.95 × 100 = $95
250 GB raw storage	S3 Standard	$3
1 TB derived storage	S3 IA	$12.50
AI Copilot (moderate use)	Bedrock Claude	$15
Lambda invocations	10M / month	$2
RDS compute (allocation)	Aurora	$50
Redis (allocation)	ElastiCache	$10
EKS compute (allocation)	EKS	$30
CloudWatch + X-Ray	Observability	$5
Data transfer	CloudFront	$8
Subtotal		**~$355**
Reserved Instance / Savings Plan discount (30%)		-$107
Net variable cost		**~$248**
Retail price ($480/seat × 5)		$2,400
Gross margin		**90%**

13.3 Reserved capacity to plan for

Resource	Monthly reservation	Cost saving
RDS Aurora (db.r6g.4xlarge reserved × 3 AZs)	$1,800 / month	35% vs on-demand
ElastiCache (cache.r6g.large reserved × 6 shards)	$580 / month	30% vs on-demand

Resource	Monthly reservation	Cost saving
EKS compute (Savings Plan, 1-year)	$4,500 / month	30% vs on-demand
EC2 GPU (P5 Capacity Block)	$18,000 / month	25% vs on-demand

14. Deployment Topology

14.1 CI/CD pipeline

```
┌──────────────────────────────────────────────────────────────────────┐
│  GitHub Actions (Phase 0 CI)                                         │
│                                                                      │
│  1. Lint (ESLint + Prettier)                                         │
│                                                                      │
│  2. Typecheck (tsc)                                                  │
│                                                                      │
│  3. Unit tests (Vitest + node:test)                                  │
│                                                                      │
│  4. Integration tests (Postgres service container)                   │
│                                                                      │
│  5. Boot smoke test (curl against running service)                   │
│                                                                      │
│  6. Build (tsc + Next.js)                                            │
│                                                                      │
│  7. Push image to ECR                                                │
│                                                                      │
│  8. Tag image with git SHA                                           │
│                                                                      │
│  9. Generate SBOM (syft)                                             │
│                                                                      │
│  10. Sign image (cosign)                                             │
│                                                                      │
└──────────────────────────────────────────────────────────────────────┘
                                    │
```

```
                                 ▼
┌──────────────────────────────────────────────────────────────────┐
│  ArgoCD (GitOps)                                                  │
│                                                                   │
│  - Watches Git repo `infra/k8s/` for ArgoCD Applications          │
│                                                                   │
│  - Auto-syncs to EKS clusters (prune + self-heal)                 │
│                                                                   │
│  - Manual approval gate for `prod` namespace                      │
│                                                                   │
│  - Argo Rollouts for canary deployments:                          │
│                                                                   │
│      setWeight: 1 → pause 5m → 5 → pause 10m → 25 → pause 15m → 50
│                                                                   │
│      → pause 20m → 100                                            │
│                                                                   │
│  - AnalysisTemplate: SLO burn rate, error rate, p95 latency        │
│                                                                   │
│  - Auto-rollback on SLO burn > 2x                                 │
│                                                                   │
└──────────────────────────────────────────────────────────────────┘
```

14.2 Argo Rollouts example (Phase 0 CI → Production)

```yaml
apiVersion: argoproj.io/v1alpha1
kind: Rollout
metadata:
  name: org-service
  namespace: sthyra-crm
spec:
  replicas: 6
  strategy:
    canary:
      canaryService: org-service-canary
      stableService: org-service-stable
      trafficRouting:
```

```yaml
        nginx:
          stableIngress: org-service
      steps:
        - setWeight: 1
        - pause: { duration: 5m }
        - setWeight: 5
        - pause: { duration: 10m }
        - analysis:
            templates:
              - templateName: org-service-slo
        - setWeight: 25
        - pause: { duration: 15m }
        - setWeight: 50
        - pause: { duration: 20m }
        - setWeight: 100
      rollbackWindow:
        revisions: 3
```

14.3 IaC layout

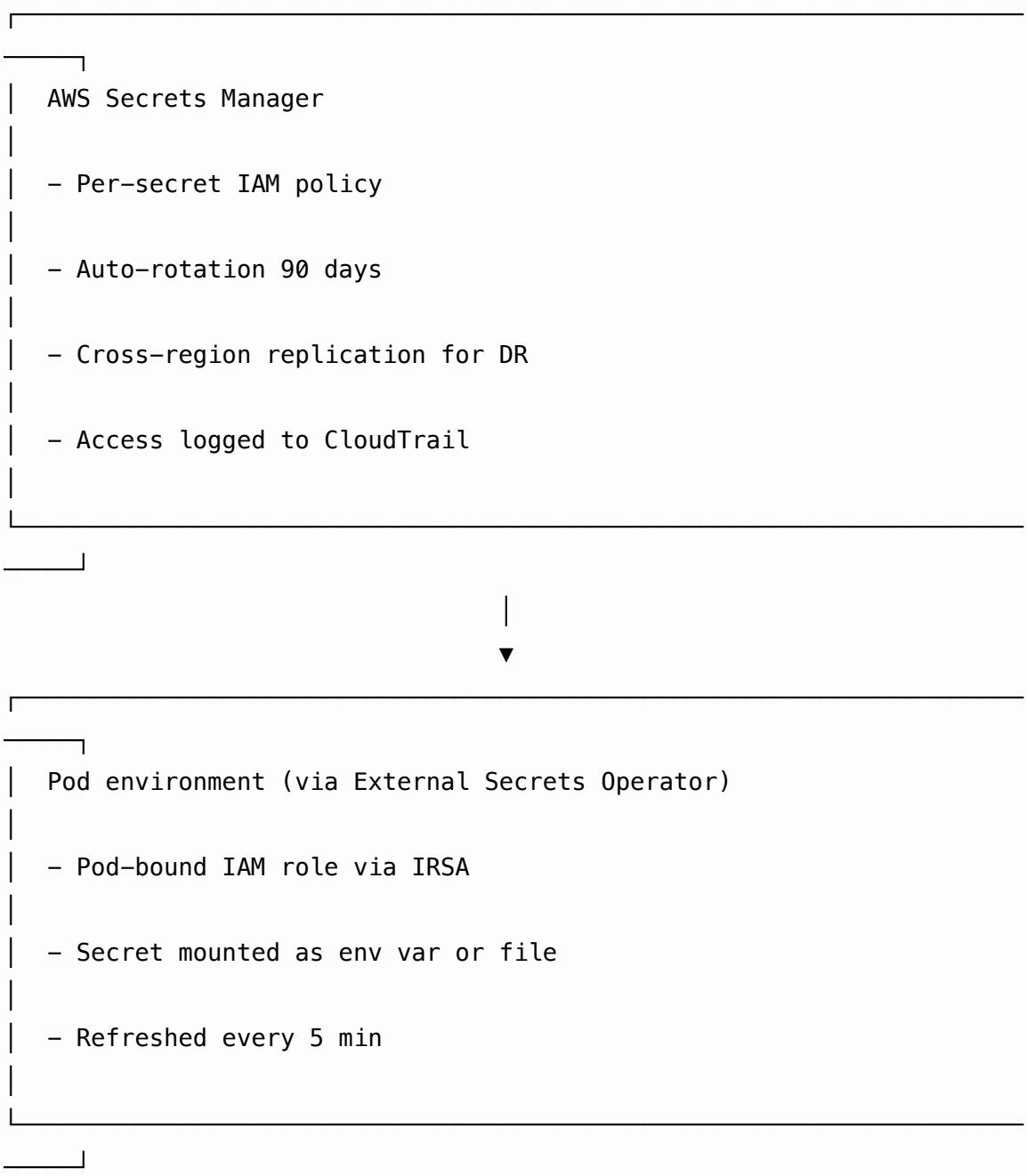
```
infra/
├── terraform/
│   ├── modules/            # Reusable modules: VPC, EKS, RDS,
ElastiCache, S3
│   ├── us-east-1/          # Region-specific stacks
│   ├── eu-west-1/
│   ├── ap-southeast-2/
│   ├── ap-northeast-1/
│   ├── ksa-central-1/
│   └── us-gov-east-1/      # FedRAMP boundary (isolated)
└── k8s/
    ├── apps/               # ArgoCD Application manifests
    ├── infra/              # Karpenter, Ingress-NGINX, cert-manager
    └── overlays/           # Kustomize overlays per environment
```

14.4 Promotion flow

```
PR opened against main
  → GitHub Actions CI runs (lint, test, typecheck, build)
```

```
  → On green: image pushed to ECR with tag "pr-{n}"
  → ArgoCD sync to dev (manual promote, sandbox)
  → Smoke test against dev
  → Manual approve → Argo Rollouts canary to staging
  → Integration tests against staging
  → Manual approve → Argo Rollouts canary to prod
  → SLO monitor watches for 1 hour
  → Auto-rollback if SLO burn > 2x
```

14.5 Secrets management



15. Risks & Trade-offs

#	Risk	Severity	Mitigation
1	EKS cluster autoscaling under GPU load spikes (large customer capture burst)	High	Karpenter MIG + multi-class GPU pool (A10G + A100 + H100) + admission quotas per tenant
2	Aurora RDS scaling bottleneck (project-membership fanout on a large org)	High	Partition by tenant_id; read replicas; pgcat pooler; hot connection pool per query
3	S3 egress cost (large customers streaming 360° tiles)	High	CloudFront Origin Shield + per-tenant signed URLs + cache-control heads + tile pyramid LOD
4	Step Functions execution cost on retry storms	Medium	DLQ-on-failure + circuit breaker; not every state retries
5	CloudWatch log volume (high-cardinality metadata)	Medium	Sampling + structured logs only; routes to S3 + OpenSearch for ad-hoc
6	Multi-region data-residency complexity	High	Per-region tenant pinning; explicit residency policy engine; fail-closed routing

#	Risk	Severity	Mitigation
7	Bedrock rate limits during Copilot burst	Medium	Per-tenant token-bucket + LLM provider fallback (Llama self-hosted)
8	Phoenix Channels WebSocket scaling for live walkthroughs	Medium	Cluster of phoenix nodes behind ALB; sticky sessions by walkthrough-id
9	SPIFFE/SPIRE operational complexity (Phase 2)	Low	Defer to Phase 2; track as Phase 2 deliverable
10	Cost attribution accuracy (CUR lag = 24h)	Low	Internal metering backed by service-level metrics for real-time cost

16. Open Decisions

These are decisions that need product / engineering / leadership input before this architecture becomes the implementation plan.

#	Decision	Options	Default	Owner
1	GPU mix for imgproc	A10G-only (cheaper) vs A100-mix (faster)	A10G-primary, A100-burst	Eng Lead
2	Vector store	pgvector (Phase 1) vs OpenSearch k-NN (Phase 2)	pgvector first, migrate if usage > 10M vectors	ML Lead

#	Decision	Options	Default	Owner
3	Mobile shell	Native (Swift + Kotlin) vs KMM	Native (matches master plan)	Mobile Lead
4	Realtime gateway tech	Elixir Phoenix vs Node + Socket.IO	Elixir Phoenix (matches master plan)	Eng Lead
5	Multi-region failover strategy	Active-active per region vs Active-passive with hot standby	Active-active for Tier 1, active-passive for GPU	Eng Lead
6	Kubernetes distribution	EKS (chosen) vs self-managed K8s	EKS	Eng Lead
7	Container runtime	containerd (default) vs gVisor	containerd for control plane, gVisor for untrusted workloads	Security Lead
8	Observability vendor	OpenSearch (chosen) vs Grafana Cloud	Self-hosted OpenSearch + Grafana	SRE Lead
9	EU AI inference	SageMaker (Mistral) vs Bedrock (Claude)	SageMaker for true EU residency	Compliance Lead
10	On-prem deployment model	EKS Anywhere vs K8s-manual	EKS Anywhere for repeatability	Eng Lead

Closing Notes

This document is the **AWS deployment target** for Sthyra CRM. It complements:

- `STHYRA-PHASE-0-REPORT.md` — what was built in Phase 0 (the foundation)

- `~/.hermes/plans/2026-08-08_090307-sthyra-crm-visual-intelligence-platform.md` — the master plan produced by the 10-agent planning pass
- `~/.hermes/plans/2026-08-08_090307-sthyra-crm-visual-intelligence-platform-APPENDIX.md` — the technical appendix with resolved cross-agent conflicts

The full system spans 13 products, ~$8.4M Year-1 budget, 12 founding hires, multi-region AWS deployment, FedRAMP Moderate authorization, 14 vendor integrations, and native mobile apps. This document specifies how it all runs on AWS. Phase 0 has shipped the foundation; the gap between this document and the working code is the execution roadmap.

End of architecture document.